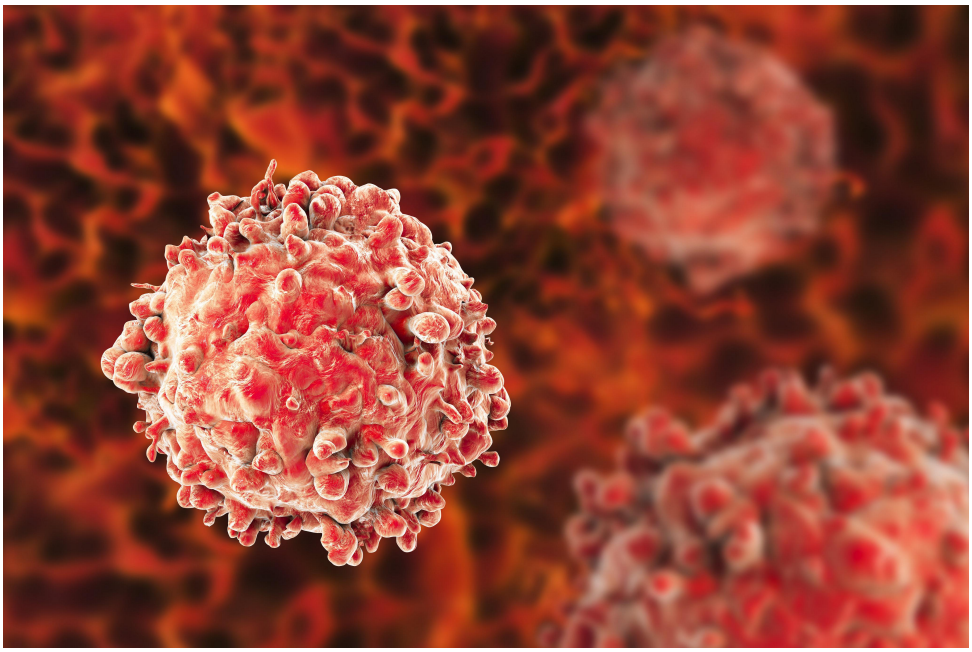
Dynamical Modeling of Chronic Myeloid Leukemia

Progression and Development of Mutations



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1 Introduction

As discussed in [1] [2], Chronic Myeloid Leukemia (CML) is a *clonal myeloproliferative disorder* of the blood, originating in the bone marrow and progressing slowly over time.

The *bone marrow* is a spongy tissue located inside long bones and certain flat bones, where immature precursor cells develop into the main blood components: red blood cells, white blood cells, and platelets. When genetic errors or mutations occur during this maturation process, a *malignant transformation* can take place, giving rise to CML.

The term *chronic* indicates that the disease evolves gradually and may remain asymptomatic for years in its initial stage.
Almost all patients with CML present tumor cells carrying the *Philadelphia chromosome*, caused by an abnormal exchange of genetic material between chromosomes 9 and 22. This chromosomal translocation generates the BCR::ABL1 *fusion gene*, which encodes an aberrant fusion protein responsible for the uncontrolled proliferation of malignant myeloid cells.

According to the World Health Organization (WHO) classification, CML progresses through three distinct phases, mainly defined by the percentage of immature cells (*blasts*) found in blood and bone marrow:

1. *Chronic phase*: blast count below 10%, with absent or mild symptoms and usually good response to therapy.
2. *Accelerated phase*: defined by one or more criteria, including a blast count between 10% and 20%, persistently high white blood cell counts despite treatment, very high or very low platelet counts, or additional chromosomal abnormalities beyond the Philadelphia chromosome. Treatment response is less favorable in this phase.
3. *Blastic (acute or blast crisis) phase*: blast count exceeds 20%. At this stage, the disease behaves more aggressively, resembling an acute leukemia.

For several years, the *standard therapy* for CML has been based on *tyrosine kinase inhibitors (TKIs)*, molecularly targeted drugs that specifically block the BCR::ABL1 fusion protein in leukemic cells while sparing healthy ones. The first of these agents, *imatinib*, revolutionized treatment and markedly improved patient survival. Imatinib therapy is generally well tolerated, with relatively mild side effects compared to traditional chemotherapies.

All the studies in [1] mainly address CML from a *clinical and therapeutic perspective*. However, to gain deeper insights into cancer behavior, it is highly valuable to combine the *clinical view* with *mathematical modeling*, in order to better understand and predict the dynamics of disease progression.

In *Section 1.1*, the mathematical model is presented and described in detail.
Section 2 focuses on the open-loop analysis, including the identification of equilibrium points, the evaluation of their stability, and the study of possible bifurcations as the parameters vary.
Section 3 introduces the control problem formulation, where both a linear (LQR) and a

nonlinear control strategy (I/O FBL) are applied.
In *Section 4*, the robustness of these two control approaches is assessed, while *Section 5* provides a quantitative comparison between them.

1.1 Mathematical model

For the generation of the mathematical model, as discussed in [1], the system is inherently *high-dimensional*. Defining such a model is crucial, since the goal is to predict the risk associated with genetic mutations. The rapid relapse of *BCR::ABL* levels observed in patients can be explained by a new conceptualization of stem cell organization. *BCR::ABL* refers to an abnormal gene sequence located on chromosome 22, also known as the *Philadelphia chromosome*. Unlike many other cancers, where the causes are often multiple and complex, in Chronic Myeloid Leukemia (CML) the presence of this specific genetic alteration represents the main cause of the disease [3].

Specifically, *stem* cells are subdivided into two compartments:

- *proliferating cells*, which are affected by imatinib treatment;
- *quiescent cells*, which remain unaffected by imatinib.

Based on these information, the following mathematical model was formulated:

$$\begin{aligned}
\frac{dT_n}{dt} &= s_n - d_n T_n - k_n T_n C_s - m q_n T_n C_r, \\
\frac{dC_s}{dt} &= k_n T_n C_s - (d_s + m\xi_n) C_s - m\epsilon_n C_r C_s, \\
\frac{dC_r}{dt} &= m q_n T_n C_r + m\epsilon_n C_r C_s - d_r C_r
\end{aligned} \tag{1}$$

where:

- T_n (*Normal Target Cells*): The population of healthy cells.
- C_s (*Sensitive CML Cells*): The population of cancer cells sensitive to the imatinib drug.
- C_r (*Resistant CML Cells*): The population of cancer cells that have developed drug resistance.
- s_n (*Source Rate*): Constant rate of normal cell (T_n) production.
- d_n (*Normal Death Rate*): Natural death rate of T_n cells.
- k_n (*Transformation Rate* $T_n \to C_s$): Rate at which T_n cells are transformed into sensitive cancer cells.
- d_s (*Sensitive Death Rate*): Natural death rate of C_s cells.
- d_r (*Resistant Death Rate*): Natural death rate of C_r cells.
- ϵ_n (*Mutation Rate* $C_s \to C_r$): Rate at which sensitive cells mutate into resistant cells.

- q_n (*Secondary Transformation Rate* $T_n \rightarrow C_r$): Rate of direct transformation of T_n into resistant cells.
- ξ_n (*Drug Efficacy Coefficient*): The parameter defining the strength of the drug's effect on C_s.

Analyzing the system of equations (1), the dynamics can be described by Figure 1 and summarized as follows:

- $\frac{dT_n}{dt}$ describes the evolution of the healthy cell population (T_n). The population increases due to the constant influx of new healthy cells, and decreases due to three main factors: natural death, transformation into sensitive leukemic cells (C_s), and direct transformation into resistant leukemic cells (C_r).
- $\frac{dC_s}{dt}$ represents the dynamics of sensitive leukemic cells (C_s). The population increases due to the conversion of healthy cells into sensitive cancer cells. Losses occur due to natural cell death and the effect of the drug, which both reduces the sensitive population and contributes to the transition toward resistant cells.
- $\frac{dC_r}{dt}$ describes the dynamics of resistant leukemic cells (C_r). The population increases through direct transformation from healthy cells and mutations from sensitive cells. It also grows indirectly due to drug effects that may favor resistance. Decreases are attributed to the natural death of resistant cells.

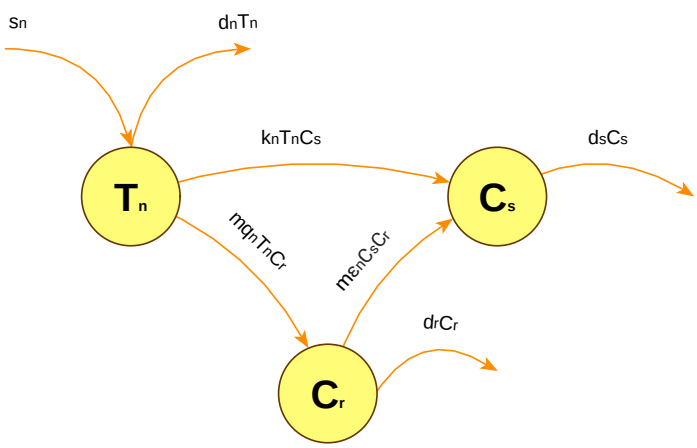


Figure 1: Dynamic behavior of the Chronic Myeloid Leukemia model, showing the evolution of healthy, sensitive, and resistant cell populations.

An important observation from [1] concerns the possibility of calibrating the model parameters using published data. However, these data have some *limitations*:

- Clinical data are continuously updated due to technological advances in detection equipment.

- Values can vary from patient to patient.

Therefore, using the exact published values may introduce *errors*.

To express the system of equations (1) in *state-space form*, it has been defined the three state variables as follows:

$$\begin{aligned} x_1 &= T_n, \\ x_2 &= C_s, \\ x_3 &= C_r. \end{aligned} \tag{2}$$

The resulting state-space representation is shown in Equation (3):

$$\begin{aligned} \dot{x}_1 &= s_n - d_n x_1 - k_n x_1 x_2 - m q_n x_1 x_3, \\ \dot{x}_2 &= k_n x_1 x_2 - (d_s + m\xi_n) x_2 - m\epsilon_n x_2 x_3, \\ \dot{x}_3 &= m q_n x_1 x_3 + m\epsilon_n x_2 x_3 - d_r x_3. \end{aligned} \tag{3}$$

It is important to highlight that, from this point onward, the case in which the medication is administered ($m = 1$) is considered. Nevertheless, the variable m is still reported for the sake of completeness.

Before concluding, it is important to report the parameters used for the evaluation of the simulations:

Parameter	Value
s_n	0.073
d_n	0.04
k_n	0.0001
d_s	0.0001
d_r	0.0001
q_n	0.0005
ϵ_n	0.0001
ξ_n	0.75
m	1

Table 1: Parameters used in the simulations of the CML model.

2 | Open-loop Analysis

As the next step, an *open-loop analysis* of the system will be carried out. This analysis aims to provide a simple and clear interpretation of the system dynamics without the presence of any control law. In particular, the following tasks will be performed on the model (3), trying to apply the theoretical concepts developed during the course:

- *Equilibrium analysis*: computation of the equilibrium points of the system.
- *Local stability*: verification of the local stability of such equilibria through linearization based on the Jacobian matrix.
- *Phase portrait*: generation of the phase portrait to visualize the qualitative behavior of trajectories in the state space.
- *Bifurcation analysis*: evaluation of possible bifurcations in the system parameters.

2.1 Equilibrium Analysis

To evaluate the equilibrium points of the mathematical model, the function $\dot{x} = f(x, u)$ must be set equal to zero. This allows for the identification of all possible combinations of states and inputs that satisfy this condition. In practice, this corresponds to setting the differential equations in (3) equal to zero:

$$\begin{aligned} s_n - d_n x_1 - k_n x_1 x_2 - m q_n x_1 x_3 &= 0 \\ k_n x_1 x_2 - (d_s + m\xi_n) x_2 - m\epsilon_n x_2 x_3 &= 0 \\ m q_n x_1 x_3 + m\epsilon_n x_2 x_3 - d_r x_3 &= 0 \end{aligned} \tag{4}$$

Through a series of algebraic manipulations, four equilibrium points were derived, as indicated in [1]. Among these, three equilibrium points are trivial, with some variables set to zero. In such cases, no interaction occurs among the cell populations, since one or more species are absent from the system. The fourth equilibrium point, instead, is non-trivial, with nonzero components. This equilibrium represents a *coexistence state*, in which all cell populations interact. They are reported in Table 2.

Table 2: Equilibrium points of the system in the variables T_n, C_s, and C_r, with $D = d_n\epsilon_n + k_n d_r - q_n d_s - q_n\xi_n$

Equilibria	T_n	C_s	C_r
E_1	$\frac{s_n}{d_n}$	0	0
E_2	$\frac{d_s+\xi_n}{k_n}$	$\frac{s_n k_n - d_n d_s - d_n \xi_n}{k_n d_s + k_n \xi_n}$	0
E_3	$\frac{d_r}{q_n}$	0	$\frac{s_n q_n - d_r d_n}{q_n d_r}$
E_4	$\frac{\epsilon_n s_n}{D}$	$\frac{d_r}{\xi_n} - \frac{q_n s_n}{D}$	$\frac{k_n s_n}{D} - \frac{d_s+\xi_n}{\epsilon_n}$

In MATLAB, the *fsolve* function was initially used to compute the equilibrium point.

However, the main limitation of this approach is that the result depends on the chosen *initial condition*, leading to the identification of only one local equilibrium point, typically the one closest to the initial guess. To overcome this issue, the `solve` function was employed instead, as it allows for the analytical and global computation of all equilibrium points of the system without requiring any initial condition.

As shown in Figure 2, all four equilibrium points are displayed for each component T_n, C_s, and C_r. Additionally, four equilibrium points exist for each section of ξ_n, reflecting the efficacy of the treatment. However, from a biological perspective, since all three components represent *cell counts*, they must be greater than or equal to zero.
In this case, *biologically* only two equilibrium points, E_1 and E_3, are meaningful, because for certain parameter values E_2 and E_4 take negative values. *Mathematically*, however, all four equilibrium points are considered.

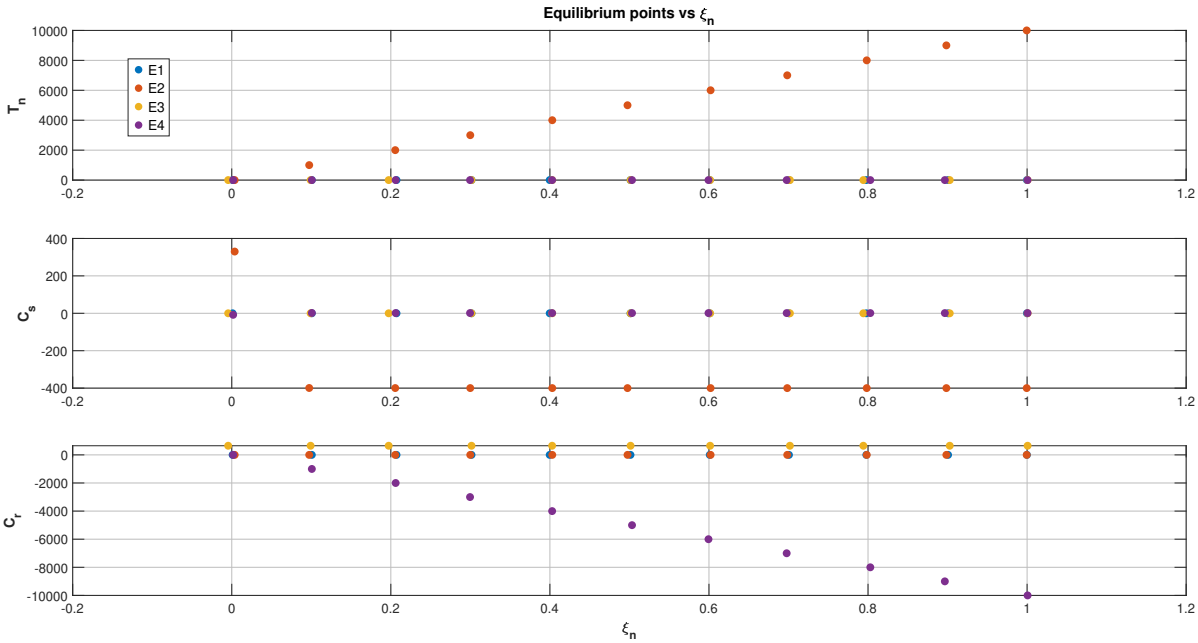


Figure 2: Equilibrium points behaviour as a function of treatment effectiveness ξ_n. The three subplots show T_n, C_s and C_r respectively.

It is important to note that the values of the equilibrium points in Figure 2 undergo a scattering along the horizontal axis, which allows for a clearer visualization and avoids overlapping of the points. Ideally, however, they are actually located on the same vertical axis.

For simplicity, the open-loop analysis was performed by varying the parameter ξ_n within the interval $[0, 1]$. Beyond this range, the system exhibits a predictable behavior, as all eigenvalues show a *monotonically* increasing trend, with no qualitative changes worth further investigation.

2.2 Equilibrium Point Stability

To verify whether the equilibrium point is *locally* stable or unstable, the system is linearized by evaluating the *Jacobian* of the nonlinear system. The Jacobian is essentially the derivative of $F(x, u)$ with respect to the state variables x, evaluated at the equilibrium point (x^*, u^*). The resulting *Jacobian matrix* is shown below:

$$J = \left.\frac{\partial F}{\partial x}\right|_{(x^*,u^*)} = \begin{bmatrix} -d_n - k_n C_s^* - m q_n C_r^* & -k_n T_n^* & -m q_n T_n^* \\ k_n C_s^* & k_n T_n^* - (d_s + m\xi_n) - m\epsilon_n C_r^* & -m\epsilon_n C_s^* \\ m q_n C_r^* & m\epsilon_n C_r^* & m q_n T_n^* + m\epsilon_n C_s^* - d_r \end{bmatrix} \tag{5}$$

The next step consists in evaluating the *characteristic polynomial* of the previously computed Jacobian matrix. This involves calculating the determinant of $(\lambda I - J)$: manually, it would be cumbersome. Therefore, the function `stability.m` was implemented and used, which directly provides, as the parameter ξ_n varies, the real and imaginary parts of the eigenvalues of the Jacobian matrix associated with the four equilibrium points . These can be visualized in Figures 3 and 4.

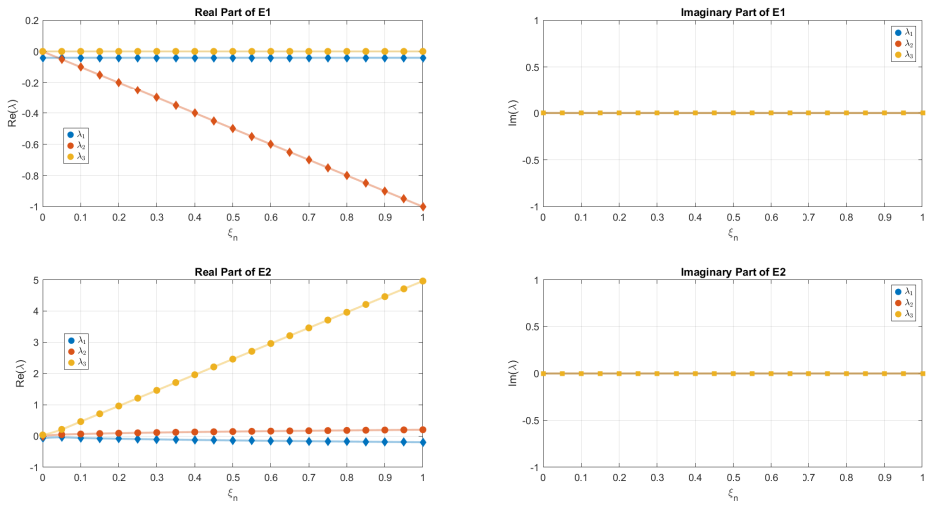


Figure 3: Real and imaginary parts of the Jacobian eigenvalues for equilibrium points E_1 and E_2 as a function of $\xi_n = [0, 1]$. Each color corresponds to a distinct eigenvalue.

From the previously cited figures, the evolution of the eigenvalues as a function of the medicine efficacy, represented by the parameter ξ_n, can be observed. It should be emphasized the meaning of the discrete markers used for the eigenvalues:

- a *circle* (○) indicates a positive eigenvalue,
- a *square* (□) suggests a zero eigenvalue,
- a *diamond* (◊) denotes a negative eigenvalue.

Based on this information, it can be concluded that only the third equilibrium point, E_3, is characterized by eigenvalues with strictly negative real parts, making E_3 a *locally stable*

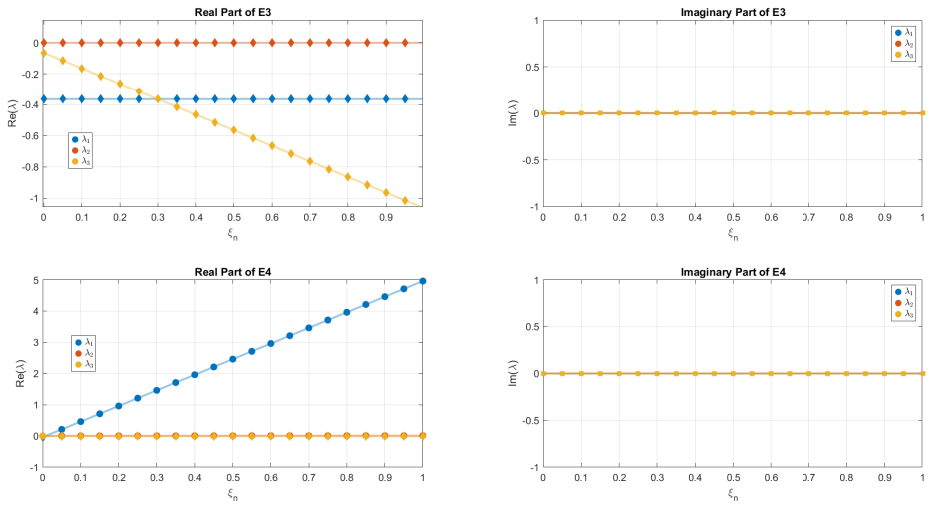


Figure 4: Real and imaginary parts of the Jacobian eigenvalues for equilibrium points E_3 and E_4 as a function of $\xi_n = [0, 1]$. Each color corresponds to a distinct eigenvalue.

equilibrium point. In contrast, the other equilibrium points exhibit at least one eigenvalue with a positive real part, and are therefore *unstable*.

Moreover, for very small values of ξ_n, a transition from negative to positive real parts is observed in Figure 5, along with the generation of nonzero imaginary parts in the eigenvalues of E_4. This behavior may represent a particular case worth further consideration.

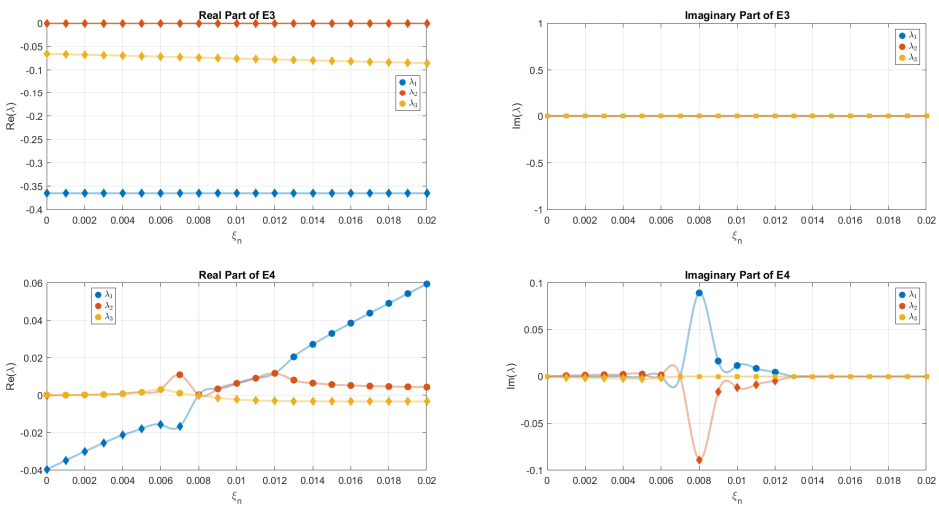


Figure 5: Real and imaginary parts of the Jacobian eigenvalues for equilibrium points E_3 and E_4 as a function of $\xi_n = [0, 0.02]$. Each color corresponds to a distinct eigenvalue.

2.3 Phase Portrait

The *phase portrait* is defined as the family of trajectories rooted at all possible initial conditions of interest within a given subset of $\mathbb{R}^n$.

Since the system under consideration is of order 3, it is possible to directly visualize the *full phase diagram*. In practice, one can select three variables and construct a three-dimensional visualization.

To represent the phase diagram of the three-dimensional system, *two main approaches* were considered, by fixing the parameter $\xi_n = 0.75$.

2.3.1 Pplane8 - Matlab Function

The *first one* involved the use of the `pplane8` tool in MATLAB, which allows the visualization of the vector field and the phase trajectories of a dynamical system in a two-dimensional plane, as a function of two state variables; in this case, it is essential to fix one of the variables to a constant value. Specifically, the third variable was set equal to the corresponding component of the selected equilibrium point, so as to analyze the system behaviour in the neighbourhood of that equilibrium.

This technique was applied only to a *subset* of the equilibrium points, due to some limitations of the `pplane8` tool. In particular, the definition of constant parameters with fractional values made it difficult to exactly select the plane corresponding to the equilibrium component. In other cases, however, it was possible to properly fix the variable and obtain a consistent and meaningful representation of the local dynamics.

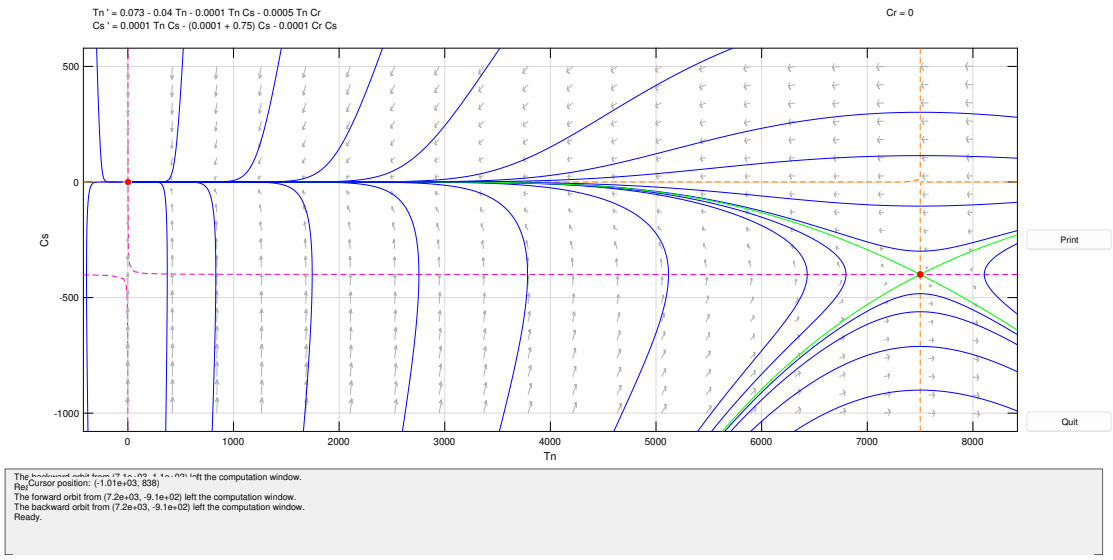


Figure 6: Phase portrait in the (T_n, C_s) plane for $C_r = 0$. The equilibrium point $E_1(1.825, 0)$ corresponds to a stable node, while $E_2(7501, -399.902)$ represents a saddle point.

From Figures 6 and 7, a fundamental yet intuitive observation can be made. For instance, considering the equilibrium point E_1, Figure 6 shows that it behaves as a stable node

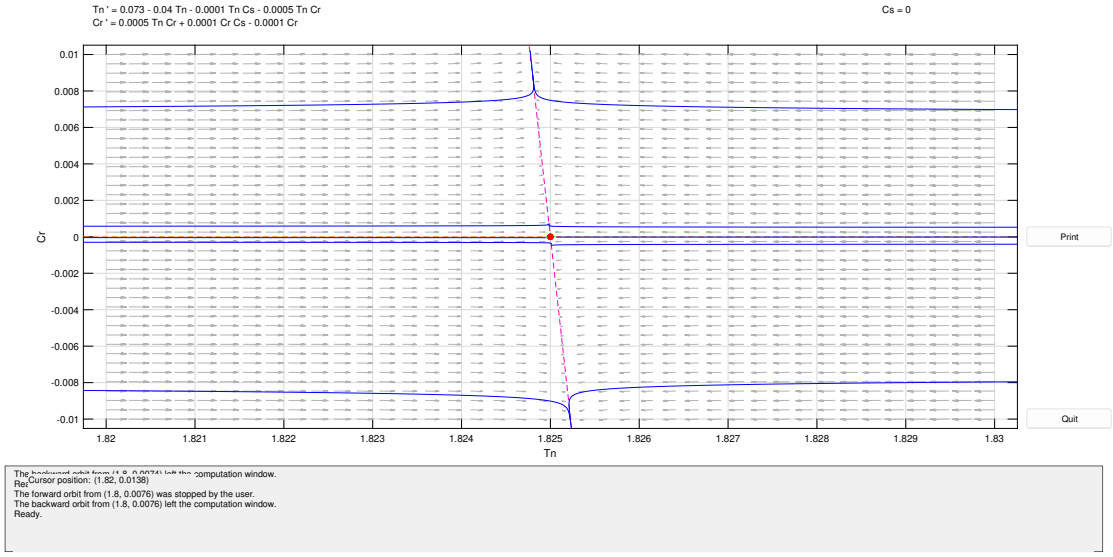


Figure 7: Phase portrait in the (T_n, C_r) plane for $C_s = 0$. The equilibrium point $E_1(1.825, 0)$ corresponds to a saddle point.

in the (T_n, C_s) plane. However, when changing the projection to the (T_n, C_r) plane, the same equilibrium no longer appears as a stable node but rather as a saddle point. Also in the (C_s, C_r) plane, the projection of the equilibrium point E_1 corresponds to a saddle point. This illustrates that, in the three-dimensional state space, the classification of the equilibrium type becomes *more complex* compared to the two-dimensional case.

Such behaviour is nevertheless expected, since, as shown in Figure 3, the eigenvalues associated with E_1 include two negative values and one positive value, confirming that the equilibrium is a *saddle-point* in the full three-dimensional system, with a one-dimensional *stable manifold*, corresponding to the eigenvector associated with the negative eigenvalue, and a two-dimensional *unstable manifold*, i.e., a curved plane tangent at the two eigenvectors associated with the two positive eigenvalues.

2.3.2 Implementation of Local and Global Phase Portrait Functions

Since the previous methodology does not allow a complete analysis of the phase diagrams for the three-dimensional system, two dedicated MATLAB functions were developed and called in the main script `open_loop_analysis.m`. One function allows the *local visualization* of the phase diagram around each of the four equilibrium points as a function of the control variable ξ_n, while the other provides a *global view* of the phase diagram, displaying all four equilibrium points in the same three-dimensional space. In both functions, it is possible to visualize the *vector fields* using gray arrows to indicate the direction and magnitude of the system's velocity, as well as the trajectories generated from initial points.
In the *local function*, initial points are chosen on a *cube centered* at the equilibrium, with side depending on the specific equilibrium point.
In the *global function*, initial points are randomly selected within the *three-dimensional cube* containing all four equilibrium points.

To indicate the *direction* of the trajectories, the starting point is marked with a *dot*, and the final point with a *star*. This representation allows for a clear visualization of both the local dynamics and the overall behaviour of the trajectories.

In the following, the results of the two MATLAB functions `phase_portraits_local` and `phase_portraits_global` are shown.

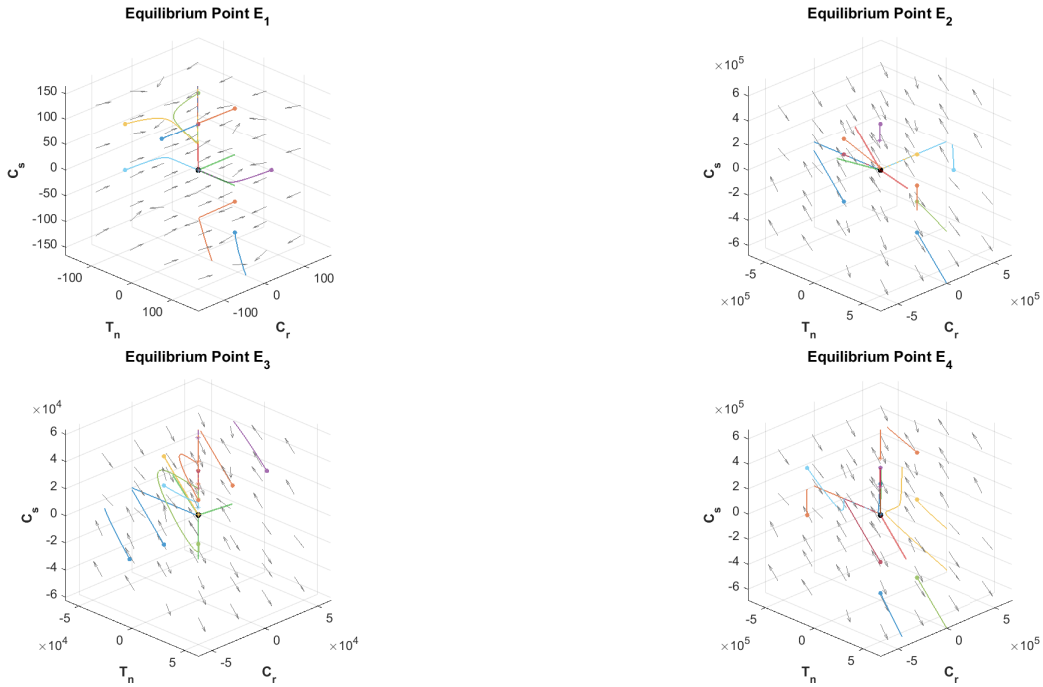


Figure 8: Local Phase Portrait with $\xi_n = 0.75$

From Figure 8, the following information can be derived. For each three-dimensional plot, a black point represents the equilibrium point. Additionally, for each equilibrium point, three segments in green or red are shown. These segments essentially represent the eigenvectors associated with the eigenvalues of the Jacobian at the equilibrium point. For example, the third equilibrium point is an *attractor*, as it is characterized by three eigenvalues with strictly negative real parts, and in this case, all three segments are green; this information also implies that E_3 is a locally asymptotically stable equilibrium point.

As explained previously, trajectories are selected within a three-dimensional cube around the equilibrium point, and each trajectory is integrated over a fixed time interval. Therefore, trajectories do not necessarily reach the equilibrium point; they are intended solely to illustrate the behavior of the system.

Regarding the equilibrium points, consider the *first point*: on the (T_n, C_r) plane, there are two eigenvalues with strictly negative real parts. Trajectories lying on this plane tend toward the equilibrium point, while trajectories along the C_s axis diverge. Similar behaviors are observed for the other equilibrium points. In particular, for the third equilibrium point almost all trajectories converge to the equilibrium point, except for those initially far away, which are influenced by the eigenvector of the first equilibrium point, as the

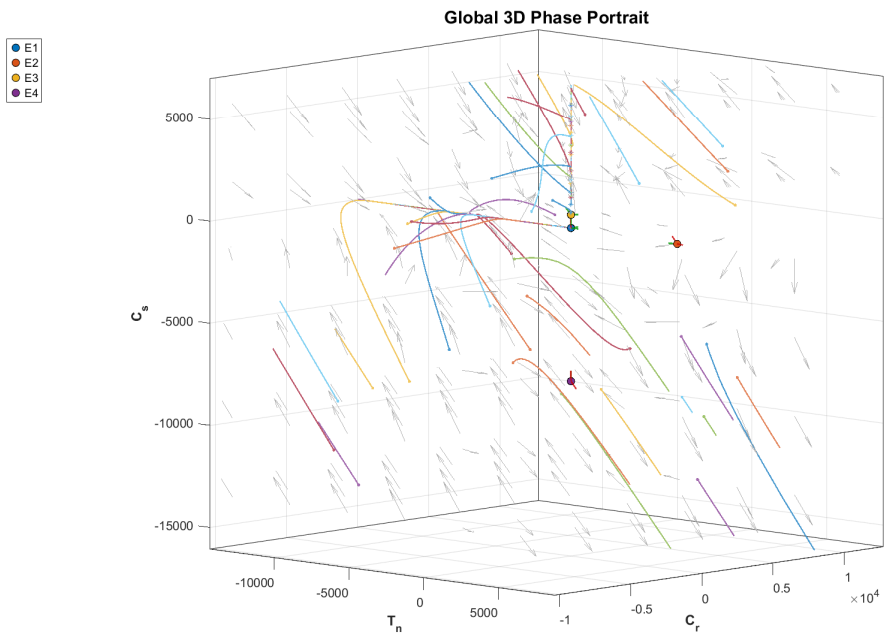


Figure 9: Global Phase Portrait with $\xi_n = 0.75$

first and third points are roughly aligned along the vertical axis, resulting in divergence.

Finally, in the fourth equilibrium point, although it is not immediately apparent, there exists an eigenvector along the vertical axis with a negative eigenvalue and, with a slight tilt, another eigenvector with a positive eigenvalue. As a result, some trajectories initially move upward before descending along the vertical axis and ultimately converge to the equilibrium point.

From Figure 9, the global phase diagram can be visualized. As can be observed, the visualization is somewhat complex, particularly because the equilibrium points are very far apart from each other. However, the general behavior can still be understood, also thanks to the representation of the vector fields in the three-dimensional space.

In general, trajectories in the upper right and lower left regions tend to move toward the upper left, eventually converging either to equilibrium point E_1 or to equilibrium point E_3. Regarding equilibrium point E_4, a divergent behavior is observed in most cases. As for equilibrium point E_2, specific cases were not evaluated; nevertheless, various simulations showed that, in most cases, the behavior is divergent due to the presence of two eigenvectors with strictly positive real parts.

2.3.3 Case Study - Analysis for ξ_n = 0.008

As discussed in Section 2.2, it was observed that for small values of the drug efficacy parameter ξ_n, the eigenvalues associated with the equilibrium point E_4 exhibit noticeable changes in their behavior. Therefore, in order to better understand what was happening, a phase portrait analysis was carried out for three different cases, each considering two

state variables at a time, with $\xi_n = 0.008$, which is precisely the value at which complex conjugate eigenvalues appear.
Observing Figures 10 and 11, which show the phase portraits in the (T_n, C_s) plane and the (C_s, C_r) plane, it can be seen that the equilibrium point $E_4 = (-182.5, 913.5, -263.5)$ in both cases represents a *saddle point*.

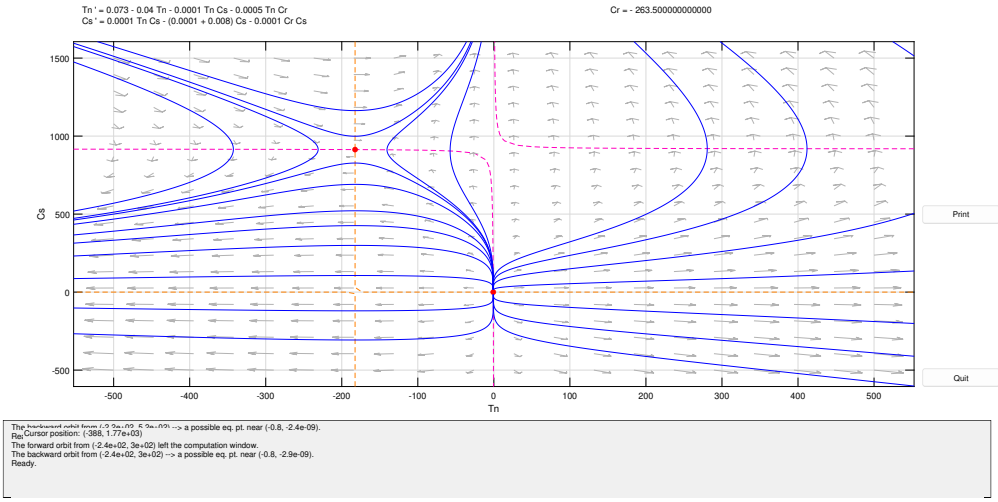


Figure 10: Phase Portrait in the (T_n, C_s) plane with $\xi_n = 0.008$

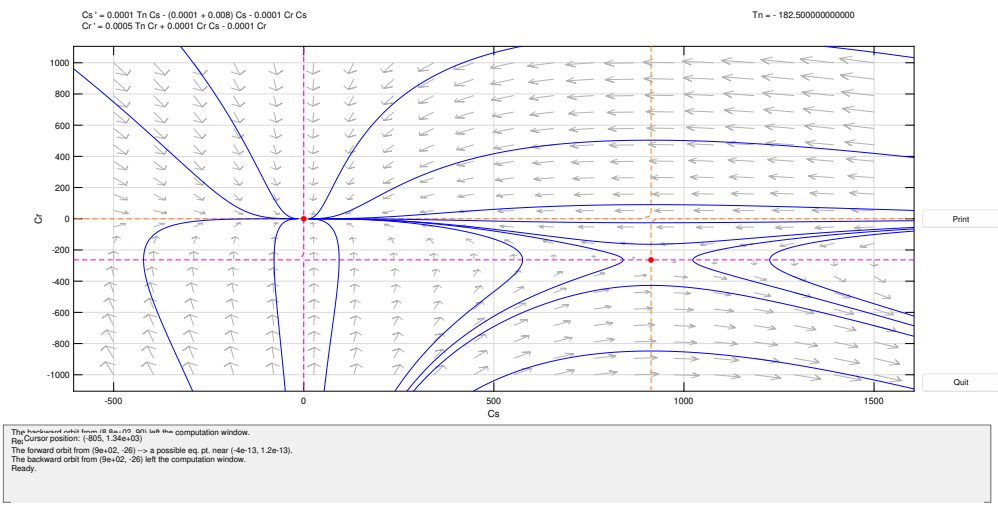


Figure 11: Phase Portrait in the (C_s, C_r) plane with $\xi_n = 0.008$.

In contrast, in the (T_n, C_r) plane, the equilibrium point E_4 appears as a *focus*, as visualized in Figure 12. By verifying with `pplane8`, the associated eigenvalues are found to be $0.0002 \pm j0.10965$. Since the real part is strictly positive, this implies an unstable focus. This type of equilibrium point can be classified as an *unstable saddle focus*, exhibiting a *tornado-like* behavior, as shown in Figure 13.

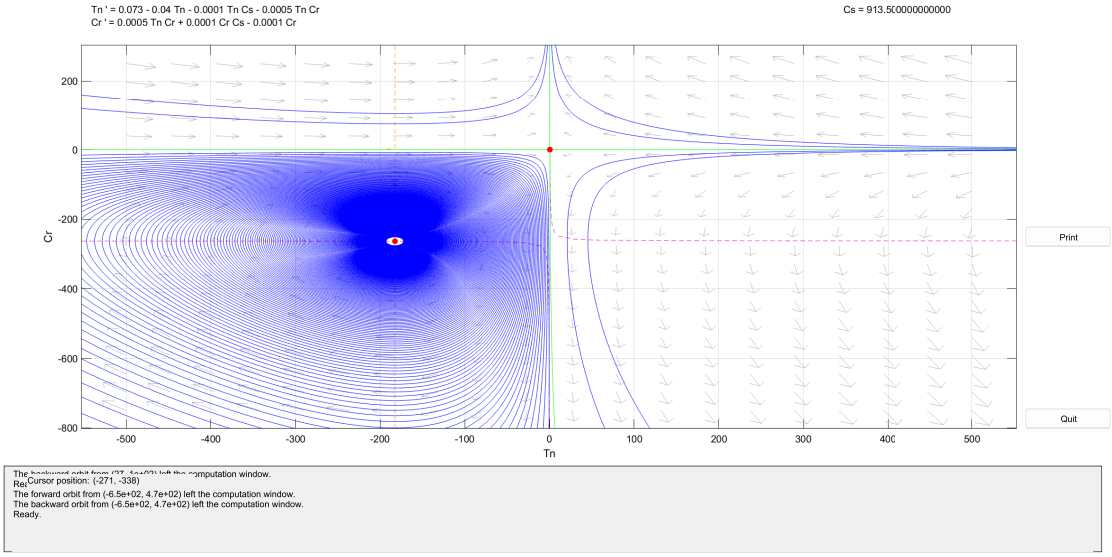


Figure 12: Phase Portrait in the (T_n, C_r) plane with $\xi_n = 0.008$.

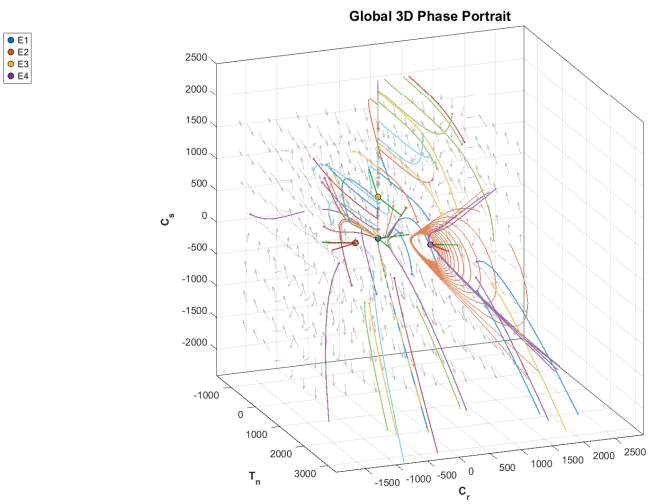


Figure 13: Global 3D Phase Portrait with $\xi_n = 0.008$.

It might be suspected from Figure 12 that there could be a *limit cycle* associated with the unstable focus. However, by applying the corollary related to the *Poincaré theorem*, in this two-dimensional case, it is essentially known that if the number N, corresponding to the number of nodes, repellers, foci, and centers, differs from the number of saddles S plus one, then no limit cycle exists.

Now, if we consider a small neighborhood around the saddle node, in that case we can be certain that the condition is not satisfied, and therefore no limit cycle exists in that neighborhood. On the other hand, if we consider the neighborhood of the focus, the condition is satisfied, and a limit cycle *could* exist. However, if we take a region that includes both components, the condition is no longer satisfied. Hence, since these are actually the only equilibrium points on this plane, we can conclude with certainty that no limit cycle exists.

2.4 Bifurcation Points

The *bifurcation points* of a nonlinear system are those points where, by varying a system *parameter*, the system loses its structural stability. During the course, particular attention has been devoted to the study of *local bifurcations*. This type of analysis has, in fact, already been partially addressed in the previous sections, where it was shown that *locally*, at a bifurcation point, equilibrium points become *non-hyperbolic*, that is, they exhibit at least one eigenvalue with zero real part.

To begin the analysis, it is necessary to visualize the evolution of the eigenvalues as a function of the parameter ξ_n. This behavior can be observed in Figure 3, 4.
Using the function `identification_range_bif_points`, a specific range of ξ_n values is considered — namely $[7.6 \times 10^{-3}, 8.3 \times 10^{-3}]$ — corresponding to the same interval described in Section 2.2 and shown in Figure 5. Within this range, zero crossings of the real part of the eigenvalues were detected, as well as the emergence of complex conjugate pairs.

Through this function, the eigenvalues of the fourth equilibrium point (the case of interest) are continuously displayed. For each interval where a possible bifurcation is detected, a corresponding message is printed in the Command Window. For simplicity, the resulting plot is not shown here, since the eigenvalue trend is identical to that already presented in Figure 5.

The function provides interesting results that require further verification, summarized as follows:

- For a parameter value of $\xi_n \approx 0.0077336$, a real eigenvalue crosses zero, changing from positive to negative. This behavior can be associated with one of three possible types of local bifurcations: *saddle-node*, *transcritical*, or *pitchfork*. A proper classification among these types must therefore be performed.
  Using `MatCont`, the equilibrium point corresponding to a parameter value of $\xi_n \approx 0.0077$ was computed and the resulting equilibrium coordinates are $(66.36, -330.82, -11.64)$.
  Starting from this equilibrium point and gradually increasing the parameter ξ_n

toward the value $\xi_n \approx 0.0077336$, `MatCont` detected a bifurcation point. From a graphical inspection of the (T_n, ξ_n) diagram, this point was identified as a *transcritical bifurcation*, as evidenced by the characteristic hook-shaped curve shown in the Figure 14.

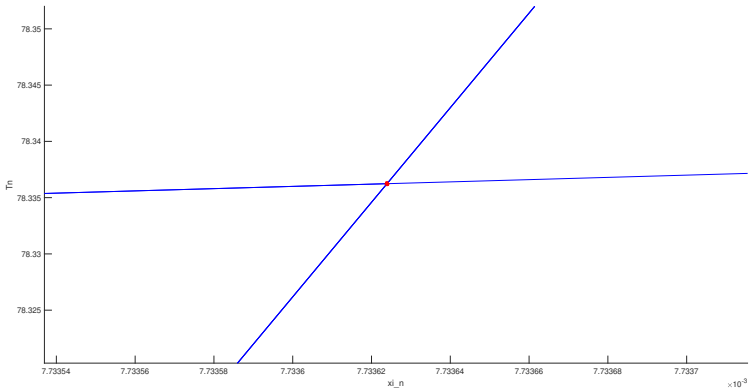
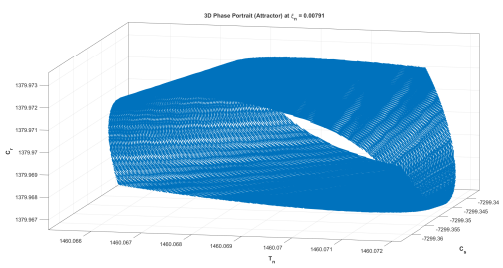
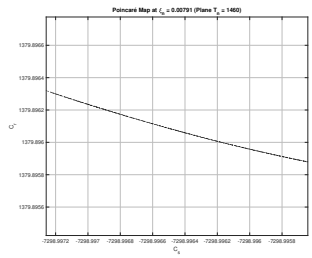


Figure 14: Transcritical bifurcation point at $\xi_n = 0.0077336$ in the (T_n, ξ_n) plane.

- In the vicinity of the parameter value $\xi_n = 0.00791$, something peculiar occurs. As shown in the Figure 15a, the trajectory does not converge to an isolated closed orbit, like a limit cycle, but rather evolves onto a *strange attractor*. Indeed, the analysis of the Poincaré map (Figure 15b) confirms that this is not a periodic attractor; instead, it exhibits a distinctive structure typical of a system exhibiting *deterministic chaos*. The numerical evidence strongly suggests that, for this particular parameter value, the system exhibits *chaotic behavior*.



(a) Chaotic behavior observed near $\xi_n = 0.00791$.



(b) Poincaré map showing the non-periodic structure of the attractor.

Figure 15: Analysis of the chaotic behavior and corresponding Poincaré map.

3 | Control Problem Formulation

In the previous sections, it has been shown how varying a *single parameter* can completely change the system's behavior, even leading to the emergence of *chaotic* dynamics. Therefore, the implementation of a *control system* to manage the evolution of leukemia in a given patient becomes essential. In this context, two types of control strategies will be applied: a *linear control* and a *nonlinear control*. The linear control will consist of linearizing the nonlinear model and applying *LQR*. As for the nonlinear control, an *Input-Output Feedback Linearization* approach will be employed. The *fundamental objective* is to ensure that, starting from certain initial conditions, the system evolves toward the *first equilibrium point* E_1, namely the state in which there are only healthy cells. This condition corresponds to the patient being essentially *cured of leukemia*. This kind of problem is called *regulation*. This means that it is not important how the different components vary over time or how they increase as time progresses, but rather that, after a certain period, the system reaches its destination - that is, the complete recovery of the patient.

Post-Simulation Analysis

Initially, a single control input u_1 was selected, corresponding to the imatinib therapy and equivalent to ξ_n. However, during the Input-Output Linearization process, it was found that the system is *underactuated*, with two outputs $y_1 = C_s$ and $y_2 = C_r$. To overcome this issue, a second control input was introduced. The control inputs u_1 (representing imatinib on C_s) and u_2 (revolutionary treatment acting on C_r) were modeled as *multiplicative efficacy rates* (x_2u_1 and x_3u_2).
This structure created critical failures for both control strategies being applied:

- For the *Input-Output Feedback Linearization*, it was found that the determinant of the decoupling matrix $E(x)$ was zero at the target equilibrium E_1, rendering the method inapplicable;
- For the *LQR controller*, linearizing the system at E_1 resulted in a null input matrix B, making the system locally uncontrollable.

These results demonstrate that any control strategy relying solely on a multiplicative input (such as ξ_n) is inherently unable to stabilize the system at the healthy equilibrium E_1. To bypass this fundamental controllability limitation, the control inputs were redefined as *additive*. From a biological perspective, this change corresponds to a shift from a *relative removal rate*, dependent on the existing cell population, to an *absolute removal rate*, comparable to a fixed-dose infusion that eliminates a specific number of cells. This additive formulation resolves both the singularity issue in the FBL scheme and the controllability failure observed in the LQR design, thereby enabling both methods to be consistently applied around the E_1 equilibrium. It is important to emphasize that the mathematical model itself remains unchanged; instead, a particular case of the original system is analyzed by imposing $\xi_n = 0$.

The new resulting state-space representation is shown in Equation (6):

$$\begin{aligned}\dot{x}_1 &= s_n - d_n x_1 - k_n x_1 x_2 - m q_n x_1 x_3,\\ \dot{x}_2 &= k_n x_1 x_2 - d_s x_2 - m \epsilon_n x_2 x_3 - \mathbf{u_1},\\ \dot{x}_3 &= m q_n x_1 x_3 + m \epsilon_n x_2 x_3 - d_r x_3 - \mathbf{u_2}.\end{aligned} \tag{6}$$

where:

- u_1: imatinib for C_s. It represents an absolute removal rate (e.g., cells/day) that acts directly on the population of sensitive cells;
- u_2: new targeted treatment for C_r. It represents an absolute removal rate that acts directly on the population of resistant cells.

In order to perform the subsequent control and robustness analyses, as well as the evaluation of the performance indices in the most fair and consistent way between the two controllers, the following *initial condition* has been chosen:

$$x_0 = [1000,\ 500,\ 250] \tag{7}$$

It is important to select an initial condition that is *biologically consistent* with the model. If an initial state is chosen outside the biologically meaningful region that is, outside the first orthant the controllers may generate invalid control laws, preventing the system from converging to the desired equilibrium point.

The *control design* will pursue the following sub-objectives:

- *Input constraints.* The drug dosage cannot be arbitrarily large, since high doses may lead to toxic effects. Therefore, it is necessary to impose a saturation on the control inputs. Each control input $u_i(t)$ must satisfy

  $$0 \leq u_i(t) \leq u_{i,\max}. \tag{8}$$

  According to article [5], the maximum daily dosage is 800 mg, but in practice this value is strongly dependent on the total number of cells present in the patient, so one can impose the maximum allowable control input as a timedependent quantity. Specifically, the upper bound on the control is given by this ratio multiplied by the number of cells:

  $$\begin{aligned}u_{1,\max}(t) &= 0.8\, C_s(t),\\ u_{2,\max}(t) &= 0.8\, C_r(t).\end{aligned} \tag{9}$$

- *Transient response speed.* It can be verified that

  $$\exists\, t \leq 180 \text{ days} \quad \text{such that} \quad C_s(t) + C_r(t) < 1. \tag{10}$$

  By consulting article [6], it can be concluded that if the patient does not reach certain thresholds of cancer cell reduction within 6 months, the therapy is considered unsuccessful. For this reason, this intermediate target was selected.

3.1 Nonlinear Control - I/O Feedback Linearization

The *Input-Output Feedback Linearization* is a control technique that computes a nonlinear feedback law to actively cancel the complex dynamics of the system, forcing the output to follow a simple and predictable linear behavior.

It is important to emphasize that, due to the aforementioned considerations, the system configuration evolved from a SISO (Single-Input Single-Output) to a MIMO (Multiple-Input Multiple-Output) structure, as a *second control input* was implemented.

For the application of the Input-Output Feedback Linearization to a MIMO system, reference is made to [4].

The *first step* consists in defining the nonlinear vector field $f(x)$ and the input matrix $g(x)$, which are reported below:

$$f(x) = \begin{bmatrix} s_n - d_n x_1 - k_n x_1 x_2 - q_n x_1 x_3 \\ k_n x_1 x_2 - d_s x_2 - \epsilon_n x_2 x_3 \\ q_n x_1 x_3 + \epsilon_n x_2 x_3 - d_r x_3 \end{bmatrix}, \quad g(x) = \begin{bmatrix} 0 & 0 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}. \tag{11}$$

Subsequently, the output equations y_1 and y_2 are differentiated until their derivatives explicitly depend on the control inputs:

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_1(x) \\ h_2(x) \end{bmatrix} = \begin{bmatrix} C_s \\ C_r \end{bmatrix}, \quad \dot{y} = \begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \end{bmatrix} = \begin{bmatrix} \dot{h}_1(x) \\ \dot{h}_2(x) \end{bmatrix} = \begin{bmatrix} \dot{C}_s(u_1) \\ \dot{C}_r(u_2) \end{bmatrix}. \tag{12}$$

The *decoupling* matrix $E(x)$ is then computed as:

$$E(x) = \begin{bmatrix} L_{g_1}(h_1) & L_{g_2}(h_1) \\ L_{g_1}(h_2) & L_{g_2}(h_2) \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}. \tag{13}$$

The feedback linearization is possible if and only if the decoupling matrix $E(x)$ is *invertible*. Evaluating its determinant, it follows that

$$\det(E(x)) = 1 \neq 0. \tag{14}$$

This condition (14) implies that the control strategy can be effectively always applied. Through this analysis, it is found that the system has two relative degrees, $r_1 = 1$ and $r_2 = 1$. Since the total relative degree, defined as the sum of the individual relative degrees ($r = r_1 + r_2$), is strictly smaller than the system order ($n = 3$), it follows that the system exhibits an *internal dynamics* of dimension one.

To verify whether this internal dynamics is *stable*, the outputs y are set to zero. This condition implies that the two state variables x_2 and x_3 must also be set to zero. As a result, only the first equation of the system remains, which can be expressed as:

$$\dot{x}_1 = s_n - d_n x_1. \tag{15}$$

Observing the internal dynamics can be simplified by restricting attention to the *zero dynamics*. For a system expressed in normal form with outputs y of relative degree r, the zero dynamics are obtained by imposing:

$$y = 0, \qquad \dot{y} = 0, \qquad \ldots, \qquad y^{(r-1)} = 0, \tag{16}$$

so that the resulting reduced flow describes the evolution of the internal (or *zero*) variables when the outputs and their time derivatives are constrained to zero.

By the *Byrnes-Isidori Theorem*, the stability properties of the zero dynamics determine the stability of the internal dynamics: if the zero dynamics are unstable, then the internal dynamics are necessarily unstable. Conversely, if the zero dynamics are *globally*, *uniformly* and *asymptotically* stable, then the internal dynamics inherit *local stability* properties under the same output constraint.

From 15, since the coefficient d_n is strictly positive, the state x_1 decays exponentially to zero; therefore, it is *asymptotically stable*, being associated with an eigenvalue with strictly negative real part. Moreover, because this is a linear differential equation with constant coefficients, local asymptotic stability implies *global asymptotic stability*. In addition, the equation is *autonomous*, meaning that it does not depend explicitly on time, and thus its stability is also *uniform*.

Consequently, since the zero dynamics are *GUAS*, this implies that the internal dynamics are locally stable. This result implies that the MIMO system is *minimum phase*, and therefore the Input-Output Feedback Linearization can be safely applied.

The relationship between the output derivatives and the input is given by:

$$\dot{y} = f_y(x) + E(x)u \tag{17}$$

where $f_y(x)$ is the vector of Lie derivatives $L_f h_i(x)$.

The *control objective* is to force the tumor populations (y_1, y_2) to decay to zero exponentially, so it has been defined a simple and stable linear dynamic v:

$$v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \dot{y}_{1\text{des}} \\ \dot{y}_{2\text{des}} \end{bmatrix} = \begin{bmatrix} -\alpha_1 y_1 \\ -\alpha_2 y_2 \end{bmatrix} = \begin{bmatrix} -\alpha_1 x_2 \\ -\alpha_2 x_3 \end{bmatrix} \tag{18}$$

where $\alpha_1 > 0$ and $\alpha_2 > 0$ are positive design gains that determine the convergence rate.

To find the control law, the actual dynamics $\dot{y}$ is set equal to the desired dynamics v and the control vector u can be solved by inverting the decoupling matrix $E(x)$, as shown in (19) and (20):

$$f_y(x) + E(x)u = v \tag{19}$$

$$u = E(x)^{-1}(v - f_y(x)) \tag{20}$$

Through algebraic manipulations, it can be derived the first and the second control law

in the following:

$$u_1(t) = \alpha_1 x_2 + k_n x_1 x_2 - d_s x_2 - \epsilon_n x_2 x_3 \tag{21}$$

$$u_2(t) = \alpha_2 x_3 + q_n x_1 x_3 - d_r x_3 + \epsilon_n x_2 x_3 \tag{22}$$

These two control laws, (21) and (22), constitute the complete feedback linearization controller. It is important to observe that both control inputs are strictly dependent on the state variables. This implies that, for the implementation of the proposed control strategy, it is necessary to employ *observers*.

3.1.1 Matlab Section

The parameters adopted for the simulation are as follows: the gains α_1 and α_2 were both chosen to be 0.5, while the range for the two control inputs, $u_1(t)$ and $u_2(t)$, is constrained between 0 and 500.

In the following, the *results* are presented as follows:

- the *Simulink block diagram* representing the structure of the Input-Output Feedback Linearization (Fig. 16);
- the *control inputs* u_1 and u_2 (Fig. 17);
- the *time evolution* of the state variables together with their tracking errors with respect to the desired components (Fig. 18);
- the *phase portrait*, which illustrates the system trajectory starting from a given initial condition x_0 (Fig. 19).

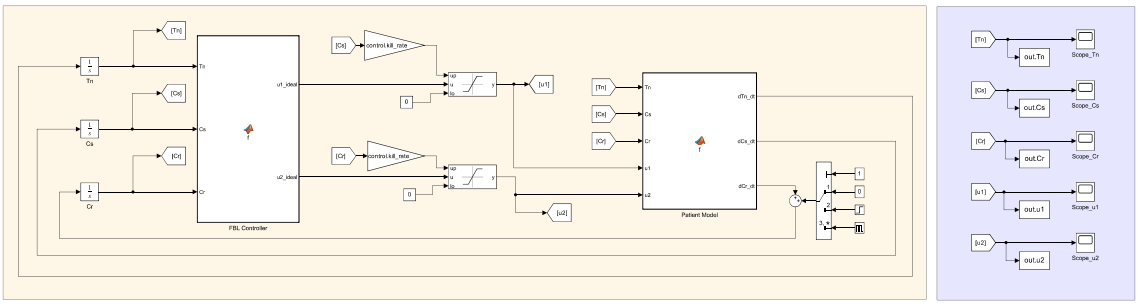


Figure 16: Simulink block diagram representing the structure of the Input-Output Feedback Linearization controller.

Figure 16 illustrates the Simulink block diagram for the implementation of the Input-Output Feedback Linearization Controller, including both the *controller component* and

the *model component*. The controller is characterized by *nominal parameters*, which are ideal and on which the control inputs are generated. The patient model, on the other hand, is characterized by the *actual parameters*, reflecting parametric uncertainty - an issue that will be discussed in *Chapter 4*.
Moreover, *dynamic saturation blocks* have been introduced to ensure that the control inputs are limited according to a maximum value, which is *time-dependent* and *proportional* to the state variables.

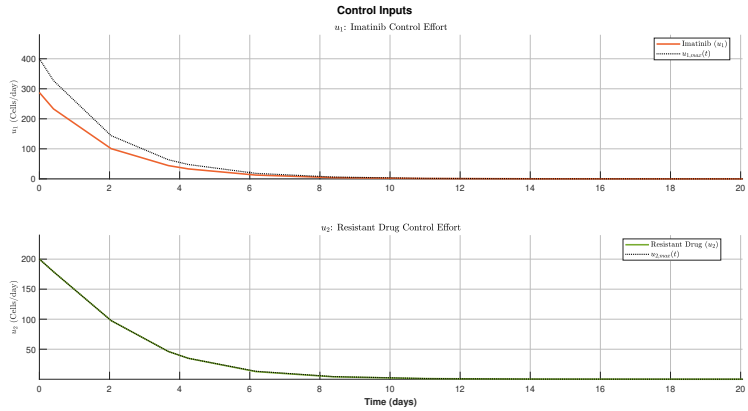


Figure 17: Time evolution of the two control inputs generated by the I/O FBL controller.

From Figure 17, it can be observed that the control input associated with the variable C_s, namely u_1, always remains below its maximum admissible limit, which is proportional to the evolution of the sensitive cell population multiplied by the coefficient 0.8. In contrast, the revolutionary control input u_2 consistently reaches saturation. This is evident from the fact that the maximum limit fully overlaps with the applied control action throughout the simulation.

In Figure 18, it is exhibit that all three state variables converge to their desired values. The *error trajectories* coincide with the evolution of the state variables over time, since, in particular, both C_s and C_r have their first equilibrium component equal to zero, while T_n reaches a value of 1.825, which is relatively small compared to the scale represented in the figure.

Furthermore, Figure 19 illustrates the *phase portrait*, starting from the initial condition x_0. It is evident that the system actually converges toward the desired equilibrium point, thereby validating the *primary goal*.

Concerning the *second sub-objective*, a separate simulation was performed in which the sum of the two state variables, x_2 and x_3, was plotted. It was observed that this sum essentially settles to one around day 14, which is significantly earlier than the required 180 days. This represents an *outstanding result*. Therefore, it can be concluded that all three objectives have been *successfully achieved*.

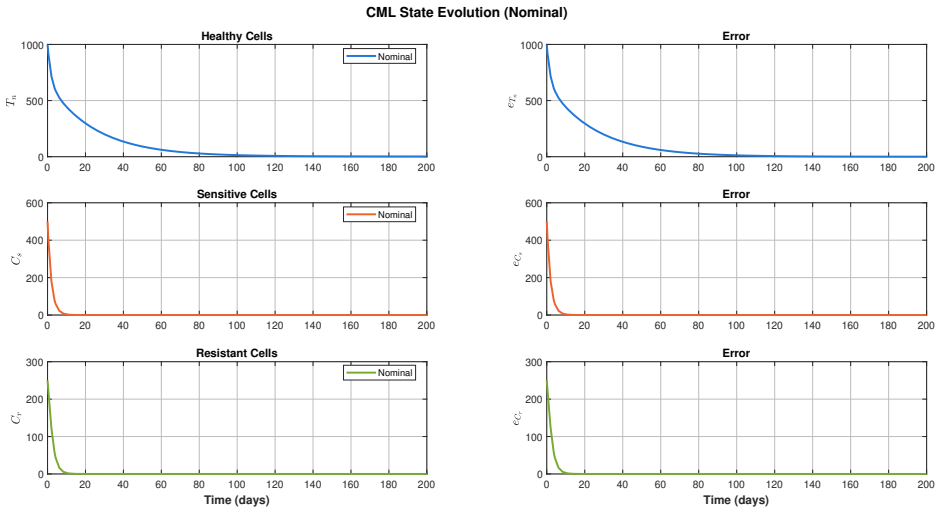


Figure 18: Time evolution of the state variables (T_n, C_s, C_r) and their tracking errors $(e_{T_n}, e_{C_s}, e_{C_r})$ with respect to the desired equilibrium values generated by the I/O FBL controller.

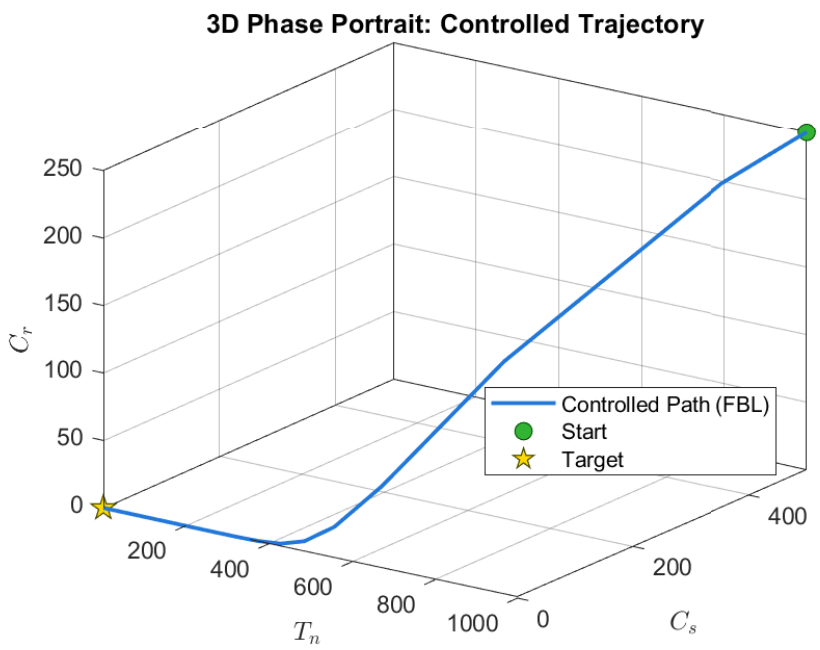


Figure 19: Phase portrait illustrating the controlled trajectory of the system, generated by the I/O FBL controller and starting from the initial condition $x_0 = (1000, 500, 250)$.

3.2 Linear Control - LQR

The *LQR control* is a linear strategy that allows assigning different levels of importance to the control inputs and the state variables. By properly tuning the weighting matrices, it is possible, for instance, to prioritize a *faster convergence* of C_s and C_r toward zero, while allowing T_n to approach its equilibrium value *more gradually*. This is achieved by adjusting the *relative weights* in the matrix Q; a similar reasoning applies to the control inputs through the matrix R.

In general, the main purpose of the `LQR_Control.m` script was to define the nonlinear system dynamics and then linearize them around the equilibrium point E_1. The *key step* consisted in verifying the rank of the *controllability matrix*, which turned out to be the critical factor motivating the introduction of additive control inputs and the removal of the multiplicative one. Indeed, it was observed that, after this modification, the system became *locally controllable* at the equilibrium point E_1. Once this verification was completed, all the necessary steps for the subsequent simulations were carried out.

3.2.1 Matlab Section

The parameters adopted for the simulation are as follows: the weighting matrices were defined as $Q = \text{diag}([0.5, 2, 2])$ and $R = \text{diag}([5, 5])$.
It was decided to assign a greater weight to C_s and C_r, since the *main objective* is to eradicate the disease, thus giving less importance to the time required to reach the corresponding equilibrium value of healthy cells (T_n). Moreover, an even higher weight was assigned to the input controls in order to avoid possible saturations within the model. For this type of controller, the same concept of input-command saturation was applied, ensuring that the upper bound was time-varying and changed proportionally with the state variables C_s and C_r.

In the following, the *results* are presented as follows:

- the *Simulink block diagram* representing the structure of LQR Control (Fig. 20);
- the *control inputs* u_1 and u_2 (Fig. 21);
- the *time evolution* of the state variables together with their tracking errors with respect to the desired components (Fig. 22);
- the *phase portrait*, which illustrates the system trajectory starting from a given initial condition x_0 (Fig. 23).

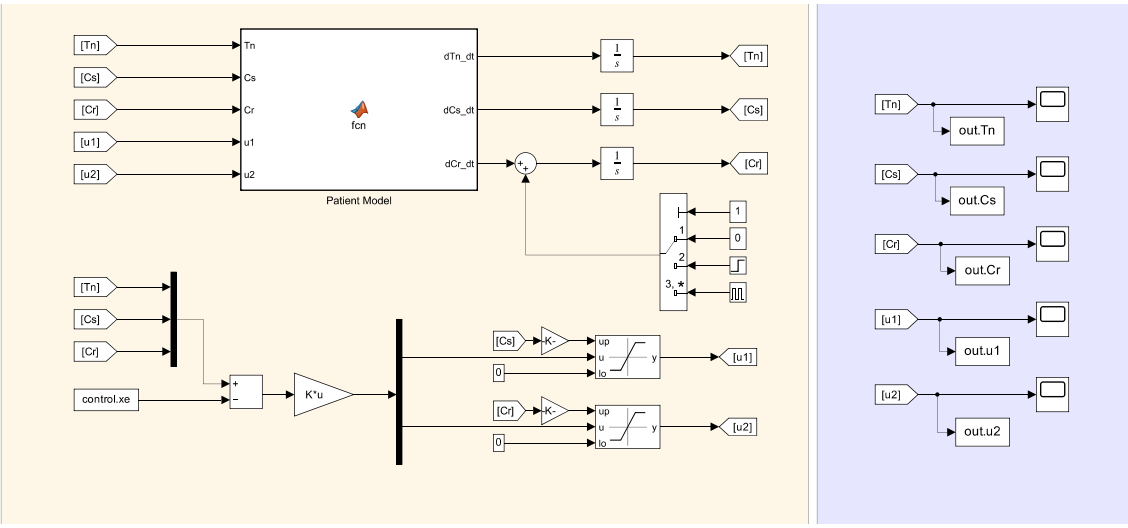


Figure 20: Simulink block diagram representing the structure of the LQR controller.

Figure 20 shows the Simulink block diagram of the LQR controller. The overall structure is essentially similar to that of the Input-Output Feedback Linearization (Figure 16). Both the block related to the dynamic saturation - whose behavior is equivalent - and the block for selecting the external disturbances are included. The only difference lies in the computation of the control inputs, which are strictly dependent on the gain K.

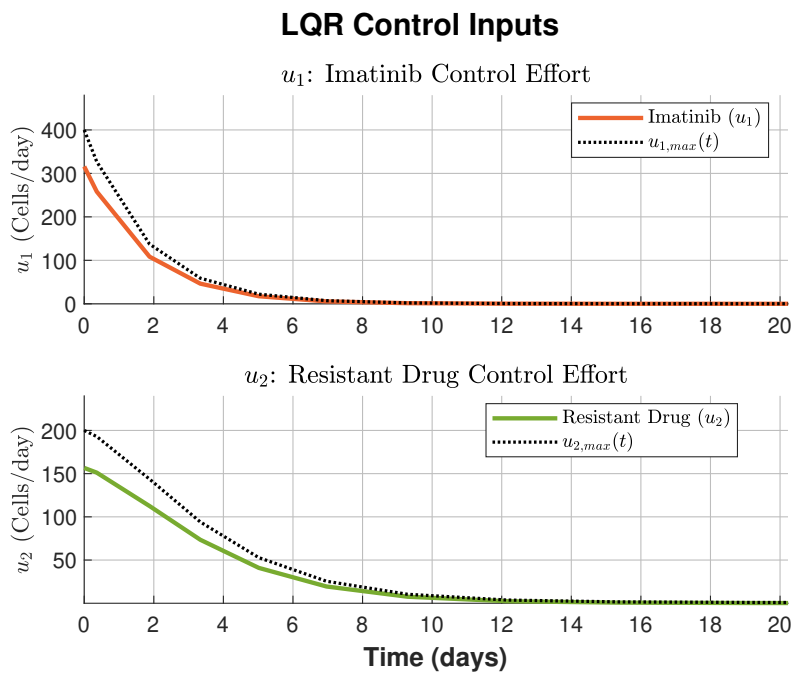


Figure 21: Time evolution of the two control inputs generated by the LQR controller.

From Figure 21, it can be observed that, overall, the control input u_1 exhibits a similar

behavior to the previous case, while the revolutionary medication u_2 reaches a *lower peak* value compared to the Input-Output Feedback Linearization. Moreover, both control inputs never reach saturation, so the *sub-objective* of remaining below the saturation limit is *fully achieved*.

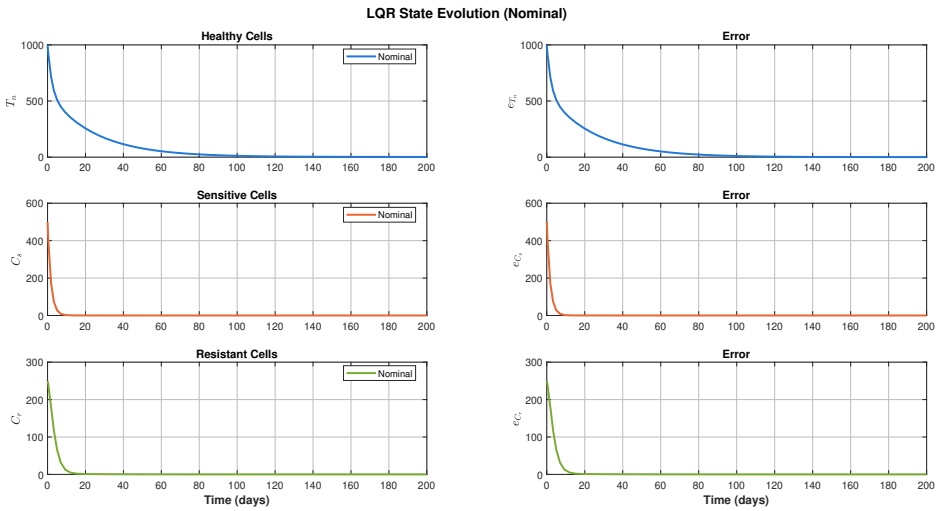


Figure 22: Time evolution of the state variables (T_n, C_s, C_r) and their tracking errors $(e_{T_n}, e_{C_s}, e_{C_r})$ with respect to the desired equilibrium values and generated by the LQR controller.

In Figure 22, it can also be observed that all three state variables converge to the equilibrium point. Initially, greater importance was assigned to the two outputs, C_s and C_r, by setting their relative weights to 100. However, in simulation, this choice caused the system and the two state variables to exhibit small oscillations during the transient phase, with an amplitude of the order of 10^{-3}, reaching the steady-state value more rapidly. It was therefore preferred to remove this negligible oscillation and allow for a slightly slower transient response instead.

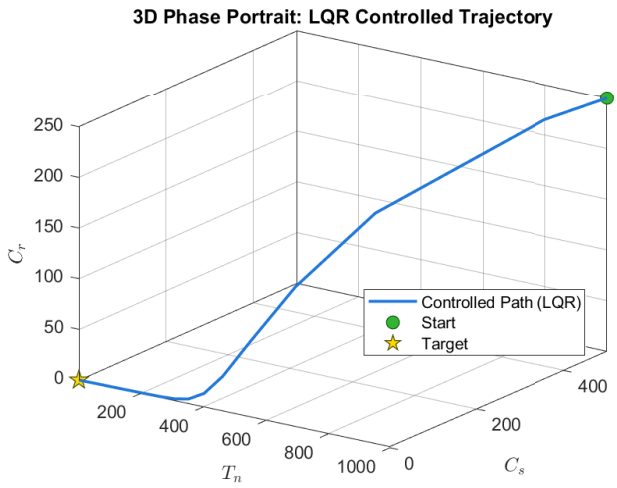


Figure 23: Phase portrait illustrating the controlled trajectory of the system starting from the initial condition $x_0 = (1000, 500, 250)$ and generated by the LQR controller.

Furthermore, Figure 23 illustrates the *phase portrait*, starting from the initial condition x_0. Also with the LQR controller application, the system actually converges toward the desired equilibrium point, thereby validating the *primary goal* for this test.

Regarding the second sub-objective, the condition (10) is immediately satisfied in 23 days. So, it can be concluded that, also with LQR controller, all three objectives have been *successfully achieved*.

4 Robustness Analysis

To perform the *robustness analysis* of the two types of controllers applied to the nonlinear model, two main techniques are adopted. The *first* consists in varying the input parameters with respect to their nominal values, by increasing them by a certain percentage. The *second* approach involves introducing an external disturbance that is not explicitly accounted for by the control laws. In both cases, the *goal* is to assess how the controllers behave when moving from an ideal case to a more realistic one, where measurement inaccuracies and modeling errors are inevitably present.

4.1 Parametric Uncertainty

As requested, an analysis was carried out by introducing uncertainty into the model parameters, in order to simulate the fact that not all patients are identical, but may exhibit different values compared to the reference patient, as reported in [1]. Several simulations were performed, increasing one parameter at a time by 50%, a value considered sufficiently representative. The parameters considered are reported in Table 1.

The simulations have shown that the system exhibits high robustness with respect to certain parameters, while it is significantly less robust with respect to others. In order to obtain a general understanding of whether the system, depending on the controller being used, is structurally robust against parametric uncertainties, a *weighted* and *normalized* performance index was computed.
The *objective* is to generate a single numerical value that quantifies the overall deviation of the three state variables from their nominal trajectories. The adopted *formula* is the following:

$$J_{w,n} = \frac{1}{3}\left(\frac{||e_{T_n}||_\infty}{T_{n,0}} + \frac{||e_{C_s}||_\infty}{C_{s,0}} + \frac{||e_{C_r}||_\infty}{C_{r,0}}\right) \cdot 100 \tag{23}$$

Table 3 summarises the obtained results: quite surprisingly, it can be observed that the Input-Output Feedback Linearization yields a value that is significantly *lower* than the score produced by the LQR controller with respect to the parameters that multiply the nonlinear terms, such as k_n, ε_n and q_n. This behavior is contrary to what one would typically expect from a controller linearized around equilibrium points. The remaining parameters, instead, show approximately the *same score*. Therefore, regarding parametric uncertainties, the Input-Output Feedback Linearization proves to be *superior*.

Table 3: Global Deviation Score: I/O FBL vs LQR

Uncertain Parameter (+50%)	Score I/O FBL (%)	Score LQR (%)
s_n	0.0341	**0.0321**
k_n	**2.4842**	2.7237
ϵ_n	**0.8108**	0.9581
q_n	**11.6553**	14.3120
d_s	0.0014	**0.0013**
d_r	**0.0023**	0.0032
d_n	3.7715	**3.3106**

Since for most parameters the deviation $J_{w,n}$ was negligible, it was decided to report only the two *main* cases.

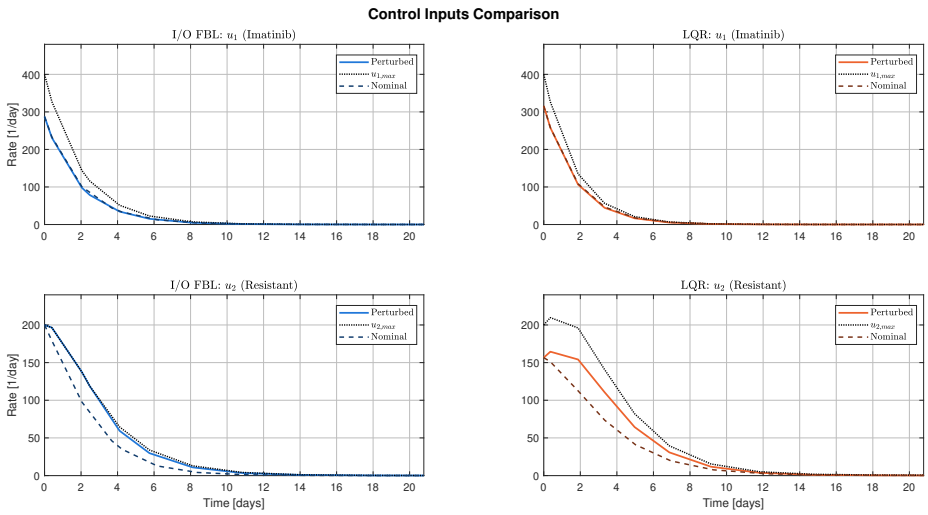


Figure 24: Comparison of control efforts (u_1, u_2) generated by I/O FBL and LQR under +50% uncertainty in the resistant proliferation rate (q_n).

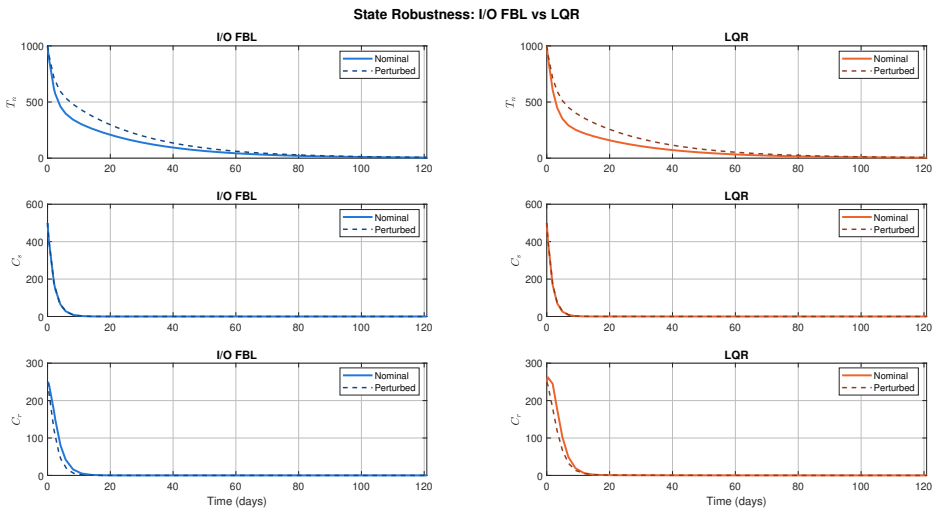


Figure 25: State robustness analysis for the critical parameter q_n (+50%) for both I/O FBL and LQR controllers.

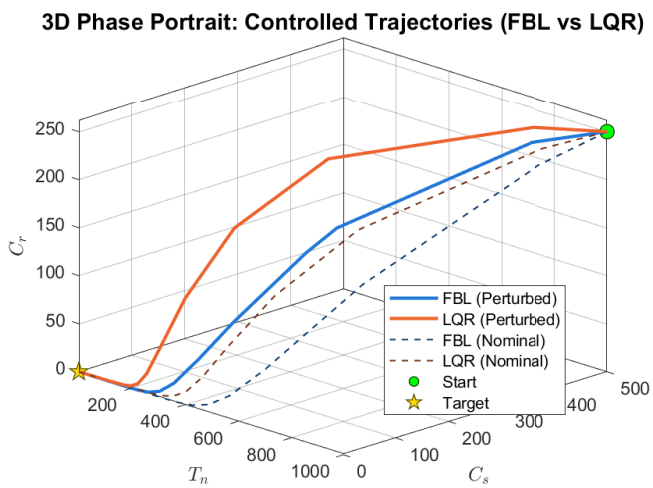


Figure 26: 3D Phase Portrait comparison under q_n uncertainty for both I/O FBL and LQR controllers.

The *first case* considered is the one with a 50% variation of the parameter q_n, which demonstrates the scenario in which the system is most affected by this parametric uncertainty. From the figure above (Figure 24), it is possible to assess the behavior of the control inputs for both controllers. With regard to the first control input u_1, the Input-Output Feedback Linearization shows a *slight deviation* from the nominal case. As for the second control input u_2, being directly influenced by the parametric uncertainty, it can be observed that in both cases it deviates from the nominal value. However, the overall trend remains almost *unchanged*.

Regarding the state variables in Figure 25, it can be clearly observed that the number of healthy cells deviates significantly from the nominal case. Indeed, this deviation becomes even more evident when examining the three-dimensional phase portrait, shown in Figure 26.

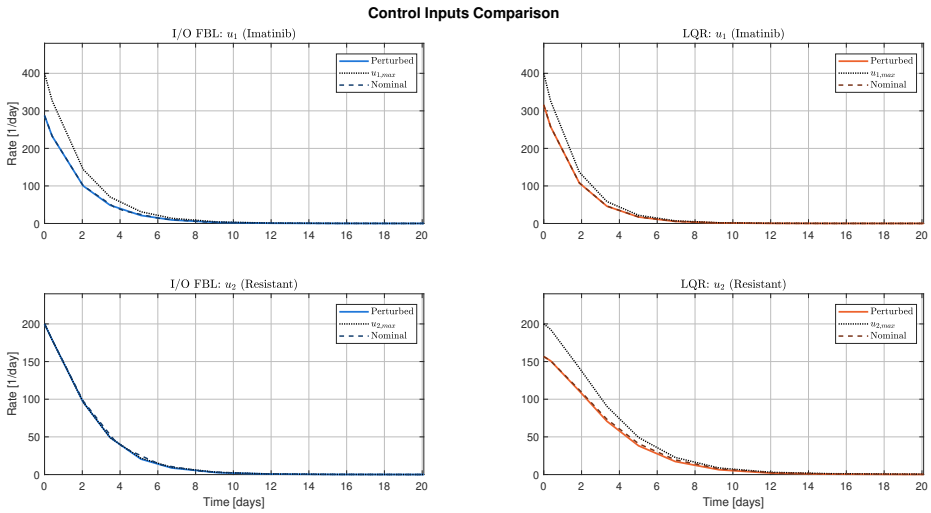


Figure 27: Comparison of control efforts (u_1, u_2) generated by I/O FBL and LQR under +50% uncertainty in the healthy cell death rate (d_n).

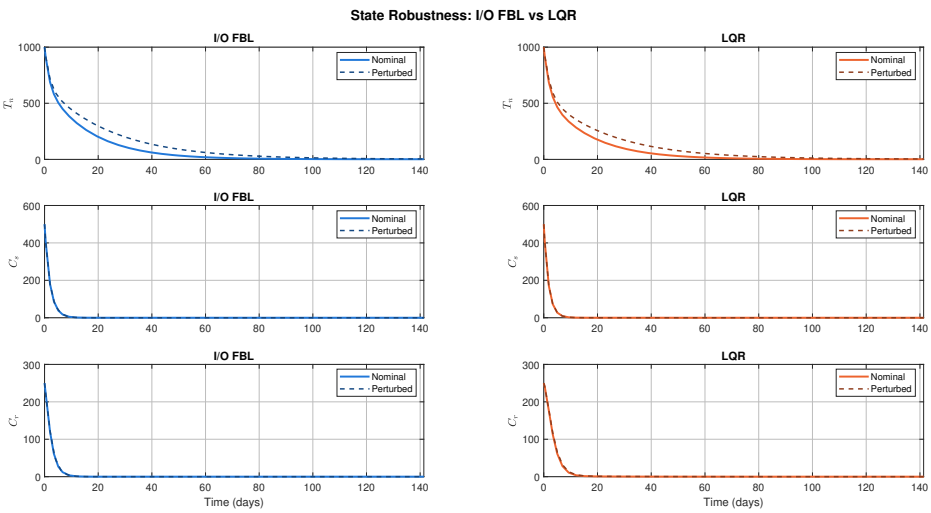


Figure 28: State evolution under structural uncertainty in d_n (+50%) for both I/O FBL and LQR controllers.

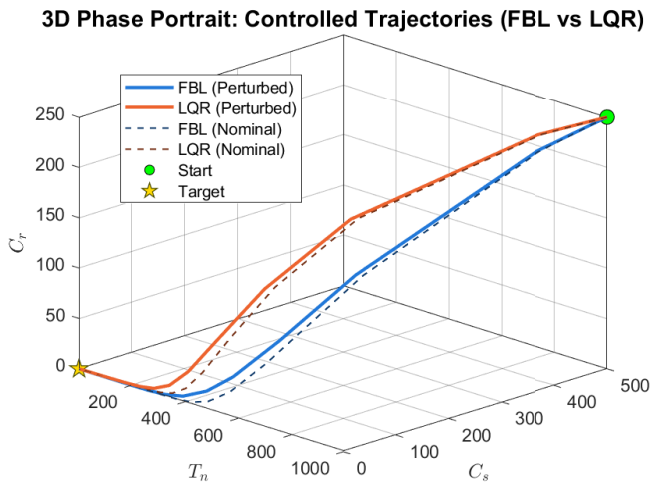


Figure 29: 3D Phase Portrait under d_n uncertainty for both I/O FBL and LQR controllers.

The *second example* considered is the case with the parametric variation of d_n, in which there is an actual change in the equilibrium point, recalled as $E_1 = [\frac{s_n}{d_n}, 0, 0]$. As a result, the system will settle at an *unintended steady-state* value. The information gathered from Figures 27, 28, and 29 indicates that the control inputs do not experience significant deviations from the nominal case. What actually changes is the first state variable, namely the number of healthy cells T_n. Since the parameter d_n is being varied, both controllers drive the system to converge not to the nominal equilibrium, but to the *real equilibrium*.

4.2 External Disturbance

For the introduction of external disturbances, a *Multiport Switch block* was implemented in Simulink control schemes. This switch is used to select and inject an external disturbance signal $d(t)$ into the system. To simulate an unmodeled growth event (such as an infection or mutation), this disturbance is added directly to the derivative of the resistant cell population, $\dot{C}_r$, effectively modifying its dynamics to:

$$\dot{C}_r = f_{y2}(x) - u_2 + d(t). \tag{24}$$

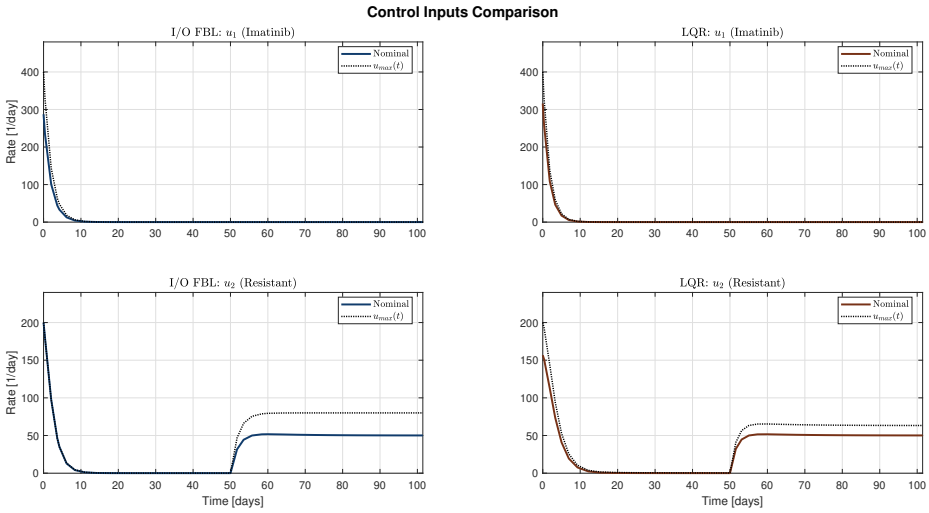
This configuration allows the selection of different input signals depending on the chosen index value. Specifically, it enables three operating *modes*:

- the *first* corresponds to an *ideal case* with a constant zero signal (no disturbance);
- the *second* introduces a *step disturbance* with amplitude 50, applied at day 50;
- the *third* employs a *pulse disturbance* with amplitude 400, period 200, width 5, and onset at day 50.

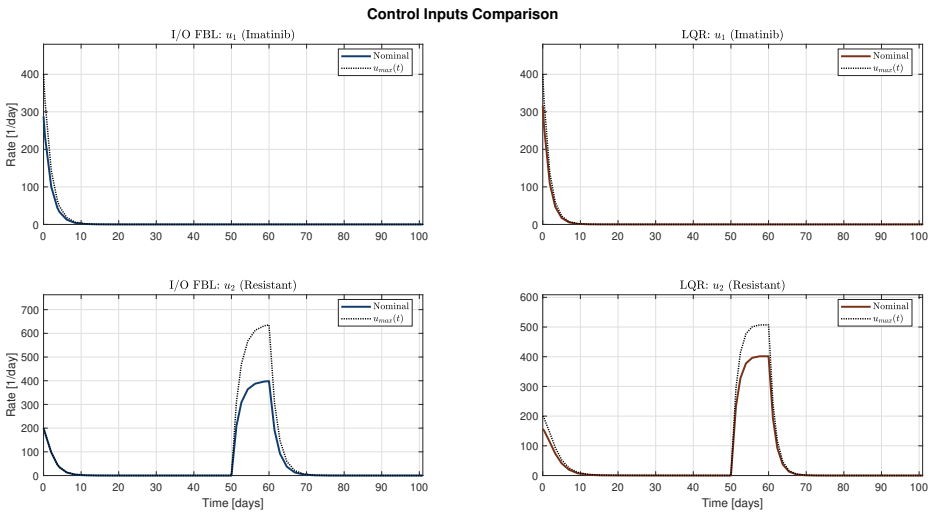
Both disturbances, in addition to making the control system *more realistic*, represent plausible physiological scenarios. For instance, the *constant disturbance* may represent a continuous generation of tumor cells resistant to Imatinib, creating a persistent load that results

in a steady-state offset. Conversely, the *pulse disturbance* serves as a model for a patient skipping a dose, resulting in a sudden increase in cell proliferation before returning to the nominal state.

For the sake of clarity, the following figures show the time evolution of the state variables, the control inputs, and the phase portraits for both controllers in the case of an applied *pulse disturbance* and in the case of an applied *step disturbance*. For this purpose, the script `plot_comparison.m` was used.



(a) Step Disturbance: amplitude 50 applied at day 50.

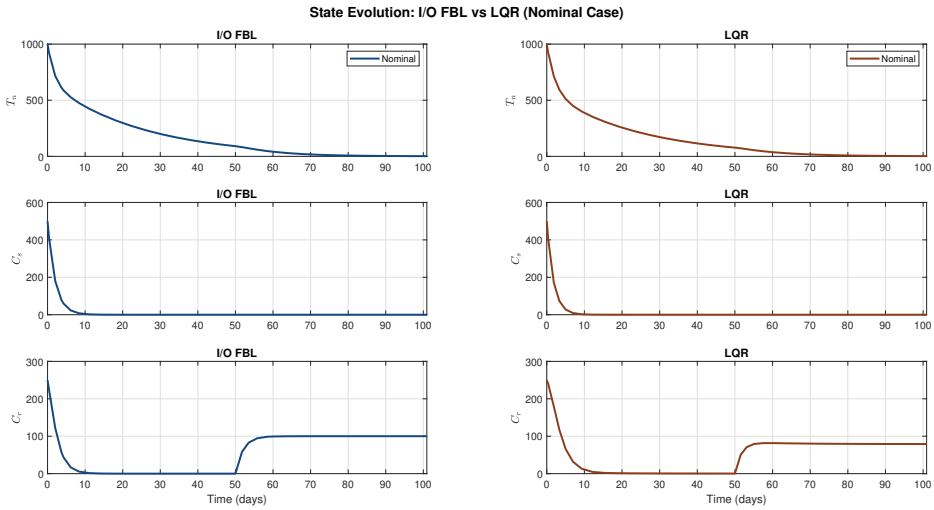


(b) Pulse Disturbance: amplitude 400 with period 200, width 5, and onset at day 50.

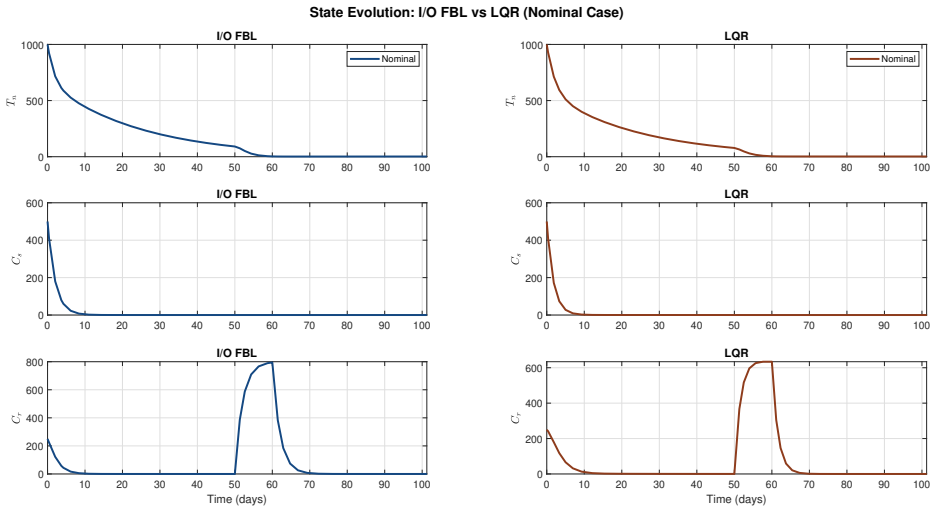
Figure 30: Comparison of Control Inputs (u_1, u_2) under Step (a) and Pulse (b) disturbances.

By comparing Figures 30a and 30b, it can be highlighted that the input control u_1, which is not directly affected by the disturbance, does not exhibit any influence from it under

either controller. Regarding the control input u_2 acting against the resistant cells, it is evident that in the case of the *step disturbance*, once day 50 is reached, the control input increases until it reaches a constant value. Both controllers behave in the same way; the *only difference* lies in the saturation level of the second control input, which is *slightly higher* for the Input-Output controller. In contrast, for the *pulse disturbance*, it can be seen that at day 50 the pulse is generated, lasting for 5 days before returning to its nominal value. Here as well, both controllers exhibit the same qualitative behavior, with the *only difference* being the magnitude of the *maximum saturation*.



(a) Step Disturbance: amplitude 50 applied at day 50.



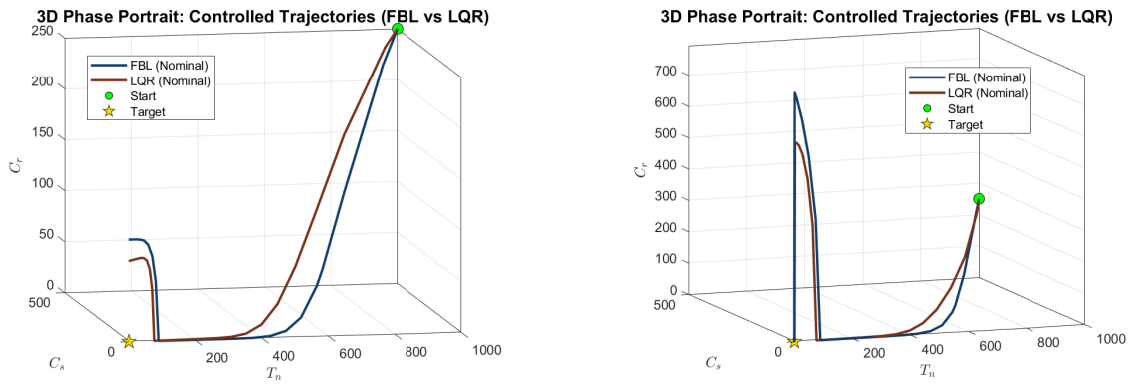
(b) Pulse Disturbance: amplitude 400 with period 200, width 5, and onset at day 50.

Figure 31: Comparison of State Trajectories (T_n, C_s, C_r) under Step (a) and Pulse (b) disturbances.

Using figures 31a and 31b, several important observations can be made. For both controllers, the second state component C_s is not affected in any way by the disturbance

input, while, as expected, the third state variable C_r is directly influenced, regardless of whether the disturbance is a step or a pulse. The state variable associated with the healthy cells, T_n, is also indirectly affected: an abrupt increase in the resistant-cell population leads, due to the nonlinearity in the first differential equation coupling T_n and C_r, to a sudden decrease in the healthy-cell population.

Another relevant aspect is that the LQR controller manages both the step and the pulse disturbances *more effectively*. In the *step case*, it drives the system toward a steady-state value closer to the equilibrium point; however, since it lacks *integral action*, the steady-state error cannot be completely eliminated in either scenario. In the *pulse case*, the amplitude of the deviation induced on the state variable C_r is *smaller* under LQR control than under Input-Output Feedback Linearization.



(a) Step Disturbance: amplitude 50 applied at day 50.

(b) Pulse Disturbance: amplitude 400 with period 200, width 5, and onset at day 50.

Figure 32: Comparison of Phase Portraits under Step (a) and Pulse (b) disturbances.

Finally, making a comparison between Figures 32a and 32b, it can be observed that, with the application of either the *step* or the *generating pulse* disturbance, the trajectories are essentially similar. However, due to the *constant disturbance*, the system cannot reach the equilibrium point because the third component continues to increase and then stabilizes at a constant value. In contrast, when the *generating pulse* is applied, in both cases the equilibrium point can be reached despite the variation in the third component.
It is evident that the LQR controller achieves *superior performance* in both scenarios.

5 | Quantitative Comparison

Once a *qualitative* and *partial quantitative analysis* have been carried out - where the system behavior is illustrated through *plots* showing its evolution, the possible malfunctioning of the control inputs, and the effects of external disturbances - it becomes essential to perform a *complete quantitative analysis* as well.

This second step aims to numerically evaluate how much the system deviates from its nominal value and to assess its robustness from a numerical perspective. Such an assessment can be achieved by introducing and analyzing specific *performance indices*.

The *performance indices* that have been considered are the following:

- *Settling time at 1%*: it represents the time required for the system to converge within 5% of the equilibrium value;
- *Steady-state error*: it measures how much each state component deviates from its steady-state value;
- *Overshoot and Undershoot*: they indicate the extent to which the system exceeds or falls below the desired response before stabilizing;
- *Cumulative Drug Dosage*: by removing the concept of static saturation and introducing *dynamic saturation* of the control inputs, it becomes much more meaningful to evaluate the amount of cumulative pharmacological load administered to the patient rather than the peak input value. In this sense, for therapies with comparable effectiveness, a lower cumulative index indicates reduced overall toxicity and therefore a more efficient control strategy:

$$C_{DD} = \int_{t_0}^{t_f} \left(u_1(\tau) + u_2(\tau)\right)\, d\tau \tag{25}$$

The *aim* of this chapter is to assess and demonstrate which controller performs better across the different cases and operating scenarios.

The following tables summarize the results obtained:

- in the *nominal* case;
- in the worst-case *parametric uncertainty* scenario (q_n);
- under an external *step* disturbance;
- under an external *pulse* disturbance.

5.1 Nominal Performance Analysis

The *nominal case* can be ideally interpreted as the scenario in which the measurements reported in Table 1 for the patient are taken as exact, and the controllers are designed strictly on the basis of these parameters. In other words, it represents a treatment tailored specifically to the reference patient, like an ideal *patient 0*.

Table 4: Performance comparison between I/O Feedback Linearization and LQR controllers in the absence of disturbances.

Metric	I/O FBL	LQR
Transient Response ($T_{a,1}$)		
Settling time of C_s [days]	22.26	86.43
Settling time of C_r [days]	18.26	126.43
Steady-State Error (e_{ss})		
Steady-State Error of C_s	1.64×10^{-28}	3.04×10^{-4}
Steady-State Error of C_r	1.20×10^{-28}	1.52×10^{-3}
State Transient Behavior		
Overshoot of C_s [%]	0.00	0.00
Overshoot of C_r [%]	0.00	0.00
Undershoot of C_s [%]	0.00	0.00
Undershoot of C_r [%]	0.00	0.00
Cumulative Drug Dosage		
C_{DD}	1109.25	1182.77

Consulting Table 4, the evolution of the patient's health states under the two control strategies in the absence of disturbances can be assessed:

- *Transient Response*: Considering the nominal patient, it is evident that the 1% *settling time* is extremely better with the Input-Output Feedback Linearization approach, with C_r component, which improves by approximately fourfold, and C_r by about sevenfold;
- *Steady-State Error*: Regarding the *steady-state error*, the components related to resistant and sensitive cells under Input-Output Feedback Linearization are significantly smaller, effectively negligible, differing from the LQR results by approximately 25 orders of magnitude, which is *remarkable*;
- *Overshoot & Undershoot*: Regarding the transient behavior, *no overshoot* or *undershoot* is observed for any component. This is evident from Figures 18 and 22, where all state trajectories are strictly decreasing and converge towards the equilibrium point;
- *Cumulative Drug Dosage*: For both controllers, the level of drug-induced toxicity is *roughly equivalent*.

Since the resulting toxicity level is essentially equivalent, it is preferable to adopt the

controller that achieves better performance in a significantly shorter time. In this regard, the Input-Output Feedback Linearization proves to be the *most advantageous option*.

5.2 Worst-Case Parametric Uncertainty (q_n)

This scenario can realistically represent a situation in which the patient has developed a *genetic alteration* that accelerates the transition from healthy cells T_n to drugresistant cells C_r under the first-line therapy with Imatinib. Alternatively, it may simply model the uncertainty arising from the measurement of this parameter, due to the *technological limitations* of the instruments used to assess the patient's clinical data.

Table 5: Performance comparison between I/O Feedback Linearization and LQR controllers under a 5% variation of q_n

Metric	I/O FBL	LQR
Transient Response ($T_{a,1}$)		
Settling time of C_s [days]	21.85	72.00
Settling time of C_r [days]	21.85	112.00
Steady-State Error (e_{ss})		
Steady-State Error of C_s	4.64×10^{-27}	2.06×10^{-4}
Steady-State Error of C_r	2.14×10^{-26}	1.03×10^{-3}
State Transient Behavior		
Overshoot of C_s [%]	0.00	0.00
Overshoot of C_r [%]	0.00	4.87
Undershoot of C_s [%]	0.00	0.00
Undershoot of C_r [%]	0.00	0.00
Cumulative Drug Dosage		
C_{DD}	1284.26	1375.81

Through Table 5, it can be observed how the two control strategies respond in the case of the uncertainty of q_n parameter:

- *Transient Response*: Despite the parametric uncertainty affecting the parameter q_n, the first controller is still able to settle within approximately 22 days, whereas the LQR requires up to 112 days. It can also be observed that, compared to the nominal case, the LQR exhibits an improvement in performance, reducing the settling time of the C_s component and the C_r component by 14 days both;
- *Steady-State Error*: The I/O Feedback Linearization controller continues to outperform the LQR. Additionally, the e_{ss} shows a slight and negligible improvement for LQR, with a reduction of approximately one unit;
- *Overshoot & Undershoot*: Regarding the undershoot, there is no variation compared to the nominal case. The only observed change is a 4.87% increase in overshoot for the third component C_r under the LQR controller, while the Input-Output Feedback Linearization controller continues to show no change;

- *Cumulative Drug Dosage*: Both controllers, more or less, show the same increase of 1.16% compared to the nominal case.

Despite the introduction of parametric uncertainty, in terms of performance, the Input-Output Feedback Linearization controller has proven to be *far superior* to the LQR.

5.3 External Disturbance Rejection

The *final case* to consider is the introduction of external disturbances into the mathematical model. In the case of a *step disturbance*, from a *biological perspective*, this could indicate the development of a *permanent change* in the patient's physiology, creating a favorable environment for resistant cells that did not previously exist, whereas in the case of a *pulse disturbance*, it could represent a group of dormant resistant cells [1] that are not initially tracked and are suddenly activated by a physiological stress.

Table 6: Performance comparison between I/O Feedback Linearization and LQR controllers under a Step disturbance

Metric	I/O FBL	LQR
Transient Response ($T_{a,1}$)		
Settling time of C_s [days]	22.26	67.63
Settling time of C_r [days]	-	-
Steady-State Error (e_{ss})		
Steady-State Error of C_s	6.59×10^{-28}	5.97×10^{-6}
Residual Error of C_r	1.00×10^{2}	7.90×10^{1}
State Transient Behavior		
Overshoot of C_s [%]	0.00	0.00
Overshoot of C_r [%]	0.00	0.00
Undershoot of C_s [%]	0.00	0.00
Undershoot of C_r [%]	0.00	0.00
Cumulative Drug Dosage		
C_{DD}	8552.37	8637.27

Through Table 6, it can be observed how the two control strategies respond in the case of a sudden and *sustained increase* in resistant cells:

- *Transient Response*: For the Input-Output Feedback Linearization controller, there is no variation in the second state variable C_s compared to the nominal case, unlike the second controller, which shows an improvement of 21.75%. It can be noted that the settling time of the third component C_r for both controllers cannot be defined, as the output tends toward a constant value far from the reference. No integral action is present to compensate for the external disturbance;
- *Residual Error*: Only the Imatinib-sensitive cells are able to reach steady-state values with a difference of at least 22 orders of magnitude, which is remarkable. For the third component C_r, the steady-state error cannot be evaluated, as the output does

not reach a reference value; instead, the residual error is considered. In this case, the second controller manages the error more effectively, yielding a value 23% smaller than that obtained with the Input-Output Feedback Linearization controller;

- *Overshoot & Undershoot*: Regarding the transient phenomena related to overshoot and undershoot, no variations are observed for either controller compared to the nominal case. In fact, all these components are equal to zero;
- *Cumulative Drug Dosage*: The first controller results in a slightly lower therapy-induced toxicity, with a reduction of 0.001% compared to the second controller, so neglibile for this case. Since the presence of C_r cells does not decrease due to the constant disturbance, it is preferable to use the controller with the lower toxicity index, namely the Input-Output Feedback Linearization controller.

Table 7: Performance comparison between I/O Feedback Linearization and LQR controllers under a Pulse disturbance

Metric	I/O FBL	LQR
Transient Response ($T_{a,1}$)		
Settling time of C_s [days]	22.26	55.67
Settling time of C_r [days]	80.13	80.73
Steady-State Error (e_{ss})		
Steady-State Error of C_s	2.78×10^{-28}	2.50×10^{-6}
Steady-State Error of C_r	5.26×10^{-16}	1.25×10^{-5}
State Transient Behavior		
Overshoot of C_s [%]	0.00	0.00
Overshoot of C_r [%]	217.83	153.62
Undershoot of C_s [%]	0.00	0.00
Undershoot of C_r [%]	0.00	0.00
Cumulative Drug Dosage		
C_{DD}	5190.81	5242.33

Table 7 illustrates the changes in performance indices following the occurrence of a 5-day pulse disturbance:

- *Transient Response*: The overall settling times of the two controllers are comparable, differing by only 0.6 days. Compared to the nominal case, the Input-Output Feedback Linearization controller shows an average deterioration of 169.41%, whereas the LQR exhibits an improvement of 35.87%.
- *Steady-State Error*: In general, there is a little degradation in the I/O Feedback Linearization compared to the nominal case, but it is considered negligible. In contrast, for the LQR, there is an improvement of 2 order of magnitude for both the variables;
- *Overshoot & Undershoot*: The only component of significance is the overshoot of C_r, which is much better managed by the LQR, showing a difference of 29.4% compared to the Input-Output Feedback Linearization controller;

- *Cumulative Drug Dosage*: The first controller results in a lower therapy-induced toxicity, with a reduction of 0.009% compared to the second controller, so negligible also for this case.

Since both the transient and steady-state responses are *comparable*, but the peak of the pulse is better managed by the second controller, the LQR linear controller is preferred for this case study.

6 | Conclusion

The *objective* of this project was to implement the *mathematical model* described in article [1]. It was initially assumed that the drug imatinib would be administered *gradually* as a function of the number of cancerous cells present. As described in the same article, this technique proves to be *insufficient* to achieve the complete recovery of the patient under consideration, who is affected by leukemia; thus, it can improve the patients health condition to some extent, but *not fully cure* it.

Therefore, the first step was to perform an *open-loop analysis* of the three-dimensional model, attempting to simulate possible scenarios by varying the drug effectiveness (ξ_n) within the interval $[0, 1]$ for simplicity. As a first step, the *equilibrium points* were computed mathematically with the aid of MATLAB program. The three-dimensional model presents *four* equilibrium points; thus, it can be stated that there are four *mathematically* valid equilibrium points, although only two of them are *biologically* meaningful. Subsequently, the stability was analyzed through the *Jacobian matrix* and by examining the real and imaginary parts of the *eigenvalues* as a function of the parameter ξ_n, and particular phenomena were observed for very small values of ξ_n. Also, the behavior of the trajectories was investigated by visualizing possible *phase portrait*s, both two-dimensional - using `pplane8` - and three-dimensional, considering the *local behavior* around each equilibrium point and the *global behavior*, in order to approximate the overall evolution of the trajectories as a function of a group of equilibrium points. Finally, the *most challenging part* was the analysis of *bifurcations*, which were evaluated within the range of very small values of ξ_n. Ultimately, a *transcritical bifurcation* was identified, along with complex behaviors suggestive of *chaos*.

Another *interesting challenge* was the implementation and design of *control actions*. The objective was to ensure that the patient, starting from a *diseased* initial condition, could eventually *recover*. From the beginning, it was known that with this type of model, it was not possible to reach the desired equilibrium point E_1. Indeed, through a series of simulations and by using the parameter ξ_n as the control input, the preliminary analyses led to the conclusion that the model was *not controllable*. Therefore, in order to avoid changing the model structure, a *new solution* was conceived: instead of administering only this type of drug, which targets sensitive cells, it was hypothesized that, in a future development, an *additional drug* could be designed to target resistant cells. By introducing these two *additive control inputs*, the mathematical model becomes *controllable*. In fact, controllability is achieved for both the Input-Output Feedback Linearization and the LQR approaches. In both cases, the equilibrium point is reached, though with different performance levels. It is also important to note that, since these two additive inputs were implemented, the parameter ξ_n was set to zero; otherwise, two inputs representing the same drug would have been applied simultaneously, which would not make *biological sense*.

Subsequently, a *robustness analysis* was carried out. The MATLAB simulations revealed that the model shows greater robustness to *parametric uncertainties*, somewhat unexpectedly, when using the Input-Output Feedback Linearization controller. Conversely, in the presence of *external disturbances*, the LQR controller achieves superior performance.

As a final consideration, the *quantitative* comparison clearly highlights a trade-off between the two strategies. Under *ideal conditions*, the Input-Output Feedback Linearization proves to be by far the *best option*, achieving much shorter settling times for cancerous cells and almost perfect steady-state accuracy, largely exceeding the extremely stringent 1% requirements.

Instead, when actual *parametric uncertainty* is introduced, in particular by examining the results in Table 3, it was observed that, compared with the nominal case, the Input-Output Feedback Linearization deviates significantly less than the LQR controller when varying the system parameters *individually*. Since the first type of controller cancels the nonlinearities exactly, this is a *surprising result*. In the few cases in which the LQR performs better, the difference with respect to I/O FBL in the Global Deviation Score is *minimal*.

On the other hand, when *external disturbances* are introduced, the LQR controller generally performs better in handling the *residual error*, reducing it by 23%. Moreover, for the maximum peak under *impulsive disturbance*, it achieves a reduction of two orders of magnitude compared with the Input-Output Feedback Linearization case.

To conclude with a broad summary, it can be stated that while the *first controller* is capable of handling *parametric uncertainties*, which may arise, for example, from measurement errors in the instruments used, with several orders of magnitude better performance, it also manages the *nominal case* effectively, representing an idealized scenario. In contrast, when *external disturbances* are present, the *linear controller* prevails, as it is able to produce lower error values with respect to the initial case, even by one order of magnitude, under the same level of treatment aggressiveness.

Disclaimer

For the purpose of simplifying the development of this project, ChatGPT was employed to perform the translation from Italian to English.

7 | Bibliography

References

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